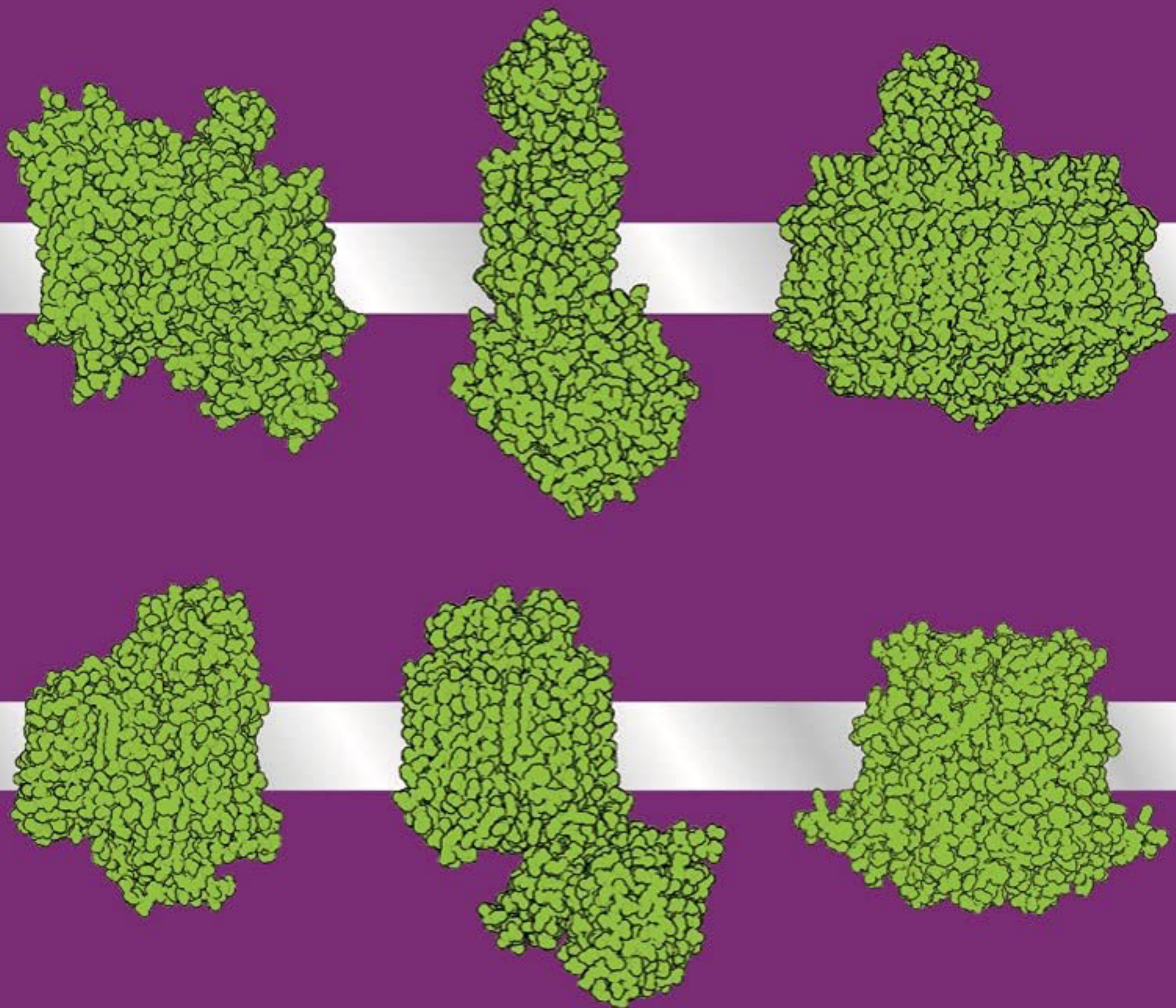


MOLECULAR BIOLOGY OF THE CELL

SEVENTH EDITION



ALBERTS HEALD JOHNSON MORGAN RAFF ROBERTS WALTER

Molecular Biology of **THE CELL**

Seventh Edition

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Bruce Alberts
Rebecca Heald
Alexander Johnson
David Morgan
Martin Raff
Keith Roberts
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With problems by
John Wilson
Tim Hunt



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Preface

Why a cell biology textbook? What is its value in a world of online resources so vast that any information you might want about cells is, in principle, freely available a few taps away?

The answer is that a textbook provides what open-ended Internet searches cannot—a curation of knowledge and an expert, accurate guide to the beauty and complexities of cells. Our book provides a narrative that leads the reader logically and progressively through the key concepts, components, and experiments in such a way that readers can build for themselves a memorable, conceptual framework for cell biology—a framework that will allow them to understand and critically evaluate the exciting rush of new discoveries. That is what we have tried to do in *Molecular Biology of the Cell* for each of its seven editions.

This edition was completed during the COVID-19 pandemic. Many of the questions that this global crisis generated are cell biological questions—including how the virus gets into our cells, how it replicates, how our immune system responds, how vaccines are developed, and how scientists produce the molecular details of virus structure. Required for the rapid development of safe and effective COVID-19 vaccines, answers to all of these questions can be found in this textbook. To make room for them, as well as for many other major recent advances in our knowledge, much previous content had to be removed.

Understanding the inner workings of cells requires more than words. Our book contains more than 1500 illustrations that create a parallel narrative, closely interwoven with the text. Each figure has been designed to highlight a key concept. The unique clarity, simplicity, and consistency of the figures across chapters, achieved by use of a common set of icon designs and colors (for example, DNA *red* and proteins *green*), enables students to scan them as chapter overviews. In this edition, important protein structures are depicted and their Protein Data Bank (PDB) codes provided; these codes link to tools on the RCSB PDB website (www.rcsb.org), where students can more fully explore the proteins that lie at the core of cell biology. In addition, more than 180 narrated movies have been produced for the book, each linked to the text to provide additional insights.

John Wilson and Tim Hunt have again contributed their distinctive and imaginative problems to help students gain a more active understanding of the text. The end-of-chapter problems emphasize experiments and quantitative approaches in order to encourage critical thinking. Their Digital Problems Book in Smartwork greatly expands on these self-assessment problems and includes data analysis and video review questions that are based on the movie links in the textbook.

Many millions of scientific papers are relevant to cell biology, and many important new ones are published daily. The challenge for textbook writers is to sort through this overwhelming wealth of information to produce a clear and accurate conceptual platform for understanding how cells work. We have aimed high, seeking primarily to support the education of cell biology students, including the next generation of bioscientists, but also to support active scientists pursuing new fundamental research and the search for practical advances to improve the human condition.

So, why read a textbook? We live in a world that presents humanity with many challenging problems related to cell biology, including declining biodiversity, climate change, food insecurity, environmental degradation, resource depletion, and animal and plant diseases. We hope that this new edition of our textbook will help the reader to better understand these problems and—for many—to contribute to solving them.

Note to the Reader

What's New in the Seventh Edition?

Every chapter in the Seventh Edition has been significantly updated with information on new discoveries in the field of cell biology. Examples of this new content include:

- Updated information on the continuing impact of human genome research, including what has been learned from sequencing hundreds of thousands of human genomes (Chapter 4), and updated coverage of tumor genomes (Chapter 20).
- New research on pathogens, diseases, and methods of combating them, including discussion of COVID-19 (Chapters 1, 5, and 23) and mRNA vaccines (Chapter 24).
- Updated research on cellular organization, including new information on biomolecular condensates (Chapters 3, 6, 7, 12, and 14) and on chromosome organization by DNA loop extrusion (Chapters 4, 7, and 17).
- Expanded coverage of new microscope technologies, including superresolution light microscopy and atomic resolution electron microscopy (Chapter 9), and new research breakthroughs from cryo-electron microscopy, such as stretch-activated Piezo channels (Chapter 11).
- New coverage of evolution, including a new discussion on the diversity of life (Chapter 1), plus updates on both human (Chapter 4) and HIV (Chapter 23) evolution.

In addition, a quarter of the book's illustrations are either completely new or significantly updated for accuracy, clarity, and visual appeal.

Finally, we are thrilled to offer online assessment, for the first time, with the Digital Problems Book in Smartwork—reimagining the classic companion text, *The Problems Book*, for twenty-first century instructors and students.

Structure of the Book

Although the chapters of this book can be read independently of one another, they are arranged in a logical sequence of five parts. The first three chapters of Part I cover elementary principles and basic biochemistry. They can serve either as an introduction for those who have not studied biochemistry or as a refresher course for those who have. Part II deals with the storage, expression, and transmission of genetic information. Part III presents the principles of the main experimental methods for investigating and analyzing cells; here, a section titled “Mathematical Analysis of Cell Function” in Chapter 8 provides an extra dimension in our understanding of cell regulation and function. Part IV describes the internal organization of the cell. Part V follows the behavior of cells in multicellular systems, starting with how cells become attached to each other and concluding with chapters on pathogens and infection and on the innate and adaptive immune systems.

End-of-Chapter Problems

A selection of problems, written by John Wilson and Tim Hunt, appears in the text at the end of each chapter. Solutions to these problems are available on the Norton Teaching Tools site.

References

A concise list of selected references is included at the end of each chapter. These are arranged in alphabetical order by author surname under the main chapter section headings. These references often include the original papers in which the most critical discoveries were first reported. The ebook also includes the DOI identifier for the references, making it easy for students to access the articles.

Glossary Terms

Throughout the book, boldface type has been used to highlight key terms at the point in a chapter where the main discussion occurs. Italic type is used to set off important terms with a lesser degree of emphasis. At the end of the book is an expanded glossary, covering all the major terms common to cell biology; it should be the first resort for a reader who encounters an unfamiliar technical word.

Website for Students

Resources for students are available at **digital.wwnorton.com/mboc7**. The complete glossary as well as a set of flashcards are available on this student website.

Nomenclature for Genes and Proteins

Each species has its own conventions for naming genes; the only common feature is that they are always set in italics. In some species (such as humans), gene names are spelled out all in capital letters; in other species (such as zebrafish), all in lowercase; in yet others (most mouse genes), with the first letter in uppercase and the rest in lowercase; or (as in *Drosophila*) with different combinations of uppercase and lowercase, according to whether the first mutant allele to be discovered produced a dominant or recessive phenotype. Conventions for naming protein products are equally varied.

This typographical chaos drives everyone crazy. Moreover, there are many occasions, especially in a book such as this, where we need to refer to a gene generically—without specifying the mouse version, the human version, the chick version, or the hippopotamus version—because the gene variants across species are all equivalent for the purposes of our discussion. What convention then should we use?

We have decided in this book to follow a uniform rule. We write all gene names with the first letter in uppercase and the rest in lowercase, and all in italics, thus: *Bazooka*, *Cdc2*, *Dishevelled*, *Egl1*. The corresponding protein, where it is named after the gene, will be written in the same way, but in roman rather than italic letters: Bazooka, Cdc2, Dishevelled, Egl1. When it is necessary to specify the organism, this can be done with a prefix to the gene name.

For completeness, we list a few further details of naming rules that we shall follow. In some instances, an added letter in the gene name is traditionally used to distinguish between genes that are related by function or evolution; for those genes, we put that letter in uppercase if it is usual to do so (*LacZ*, *RecA*, *HoxA4*). Proteins are more of a problem. Many of them have names in their own right, assigned to them before the gene was named. Such protein names take many forms, although most of them traditionally begin with a lowercase letter (actin, hemoglobin, catalase); others are acronyms (such as GFP, for green fluorescent protein, or BMP4, for bone morphogenetic protein 4). To force all such protein names into a uniform style would do too much violence to established usages, and we shall simply write them in the traditional way. For the corresponding gene names in all these cases, we shall nevertheless follow our standard rule: *Actin*, *Hemoglobin*, *Catalase*, *Bmp4*, *Gfp*.

For those who wish to know them, the table shows some of the official conventions for individual species—conventions that we shall mostly violate in this book, in the manner shown.

Organism	Species-specific convention		Unified convention used in this book	
	Gene	Protein	Gene	Protein
Mouse	<i>Hoxa4</i>	Hoxa4	<i>HoxA4</i>	HoxA4
	<i>Bmp4</i>	BMP4	<i>Bmp4</i>	BMP4
	<i>integrin α-1, Itgα1</i>	integrin α 1	<i>Integrin α1, Itgα1</i>	integrin α 1
Human	<i>HOXA4</i>	HOXA4	<i>HoxA4</i>	HoxA4
Zebrafish	<i>cyclops, cyc</i>	Cyclops, Cyc	<i>Cyclops, Cyc</i>	Cyclops, Cyc
<i>Caenorhabditis</i>	<i>unc-6</i>	UNC-6	<i>Unc6</i>	Unc6
<i>Drosophila</i>	<i>sevenless, sev</i> (named after recessive phenotype)	Sevenless, SEV	<i>Sevenless, Sev</i>	Sevenless, Sev
	<i>Deformed, Dfd</i> (named after dominant mutant phenotype)	Deformed, DFD	<i>Deformed, Dfd</i>	Deformed, Dfd
Yeast				
<i>Saccharomyces cerevisiae</i> (budding yeast)	<i>CDC28</i>	Cdc28, Cdc28p	<i>Cdc28</i>	Cdc28
<i>Schizosaccharomyces pombe</i> (fission yeast)	<i>Cdc2</i>	Cdc2, Cdc2p	<i>Cdc2</i>	Cdc2
<i>Arabidopsis</i>	<i>GAI</i>	GAI	<i>Gai</i>	GAI
<i>Escherichia coli</i>	<i>uvrA</i>	UvrA	<i>UvrA</i>	UvrA

Resources for Instructors

digital.wwnorton.com/mboc7

Designed to enrich the classroom experience, Instructor Resources are available at digital.wwnorton.com/mboc7. Adopting instructors can obtain access to the site from their sales representative, who can be identified by visiting [wwnorton.com/educator](https://www.wwnorton.com/educator) and clicking the “Find My Rep” button.

The Digital Problems Book in Smartwork

For the first time, the popular print supplement *Molecular Biology of the Cell: The Problems Book* is now available in Smartwork. Easier for instructors to assign and more helpful to students because of each question’s pedagogical scaffolding, the Digital Problems Book in Smartwork features the questions authored by Tim Hunt and John Wilson adapted for digital delivery. An enormous library of almost 3500 questions that include critical thinking questions, data analysis questions, and animation and video questions, allows instructors to deliver the exact type of assessment that their students need. The Digital Problems Book in Smartwork comes at no additional cost with all new copies of *Molecular Biology of the Cell*.

Question Detail

ODD ENZYME KINETICS FOR O⁶-METHYLGUANINE REPAIR IN DNA [BLOOM'S 4] [ART]

1st attempt

See Hint

The alkylation repair system in bacteria removes the methyl group from O⁶-methylguanine, converting it to guanine and preventing mutation. The enzyme mechanism is somewhat peculiar. The kinetics of removal were studied by incubating 1.25, 2.50, or 5.00 ng of the pure enzyme with DNA containing ³H-O⁶-methylguanine. At various times, samples were taken, and the DNA was analyzed to determine how much of the mutagenic base remained (see the figure). When the experiment was repeated at 5°C instead of 37°C, the initial rates of removal were slower, but the same end points were achieved.

What, if anything, is peculiar about the kinetics of removal of the methyl group from the O⁶-methylguanine?

Time (minutes)	1.25 ng (% remaining)	2.50 ng (% remaining)	5.00 ng (% remaining)
0	100	100	100
1	85	60	10
2	78	55	5
4	75	52	2
6	75	50	1

Choose one:

- ☐ A. One would expect the extent of reaction to increase with increasing enzyme concentration, as seen here.
- ☐ B. It is strange that removal of the methyl groups stops at a plateau that depends on enzyme concentration.
- ☐ C. The extent of removal does not change with temperature, which is unusual for enzyme-catalyzed reactions.
- ☐ D. The rate of removal of methyl groups increases with increasing enzyme concentration, as expected.

SUBMIT ANSWER

Norton Teaching Tools

The Norton Teaching Tools site for *Molecular Biology of the Cell* provides creative and engaging resources to refresh a syllabus or to design a new one. Dynamic, experienced instructors have created primary literature suggestions, active learning activities, lecture PowerPoint files, descriptions of all of the animations and videos, and much more. All of the teaching tools are aligned with chapter topics and organized by activity type, making it easily sortable. The site also features tips for assigning Norton's digital learning tools and addressing the most common course challenges.

Animations and Videos

Under the authorial direction of Michele M. McDonough and Thomas A. Volpe, both of Northwestern University, the animations and video library has been thoroughly updated and expanded. The more than 180 animations and videos are integrated into the ebook and also available to students and instructors at digital .wwnorton.com/mboc7. Instructors can view descriptions of each on the Norton Teaching Tools site.

Norton Ebook

The purchase of any new print copy of the Seventh Edition of *Molecular Biology of the Cell* includes access to the Norton Ebook version of the text at no additional cost. The Norton Ebook can be purchased as an affordable stand-alone option that provides an active reading experience, enabling students to take notes, bookmark, search, highlight, and read offline. All of the videos and animations appear directly in the ebook, and instructors can add notes that students can see as they are reading the text.

Art of *Molecular Biology of the Cell*, Seventh Edition

The images from the book are available in two convenient formats: PowerPoint and JPEG, and in both labeled and unlabeled versions.

Figure-integrated Lecture Outlines

The section headings, concept headings, and figures from the text have been integrated into PowerPoint presentations and can be customized. For example, the content of these presentations can be combined with videos, questions from the book, or activities in the Norton Teaching Tools site, in order to create unique lectures that facilitate interactive learning.

Test Bank

Updated for the Seventh Edition, the test bank includes a variety of question formats: multiple choice, short answer, fill-in-the-blank, true-false, and matching. The test bank was created with the philosophy that a good exam should require students to reflect upon and integrate information as a part of a sound understanding. Questions are classified by section and difficulty, making it easy to construct tests and quizzes. The test bank question library includes about 70 questions per chapter, ensuring instructors can find the right questions for their exams. It will be delivered through Norton Testmaker, which brings the high-quality questions in the test bank online. Create assessments for your course without downloading files or installing specialized software, customize test bank questions, and easily export your tests to Microsoft Word or Common Cartridge files for your learning management system (LMS).

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Brief Contents

PART I	INTRODUCTION TO THE CELL	
Chapter 1	Cells, Genomes, and the Diversity of Life	1
Chapter 2	Cell Chemistry and Bioenergetics	49
Chapter 3	Proteins	115
PART II	BASIC GENETIC MECHANISMS	
Chapter 4	DNA, Chromosomes, and Genomes	183
Chapter 5	DNA Replication, Repair, and Recombination	253
Chapter 6	How Cells Read the Genome: From DNA to Protein	321
Chapter 7	Control of Gene Expression	397
PART III	WAYS OF WORKING WITH CELLS	
Chapter 8	Analyzing Cells, Molecules, and Systems	475
Chapter 9	Visualizing Cells and Their Molecules	563
PART IV	INTERNAL ORGANIZATION OF THE CELL	
Chapter 10	Membrane Structure	603
Chapter 11	Small-Molecule Transport and Electrical Properties of Membranes	637
Chapter 12	Intracellular Organization and Protein Sorting	683
Chapter 13	Intracellular Membrane Traffic	749
Chapter 14	Energy Conversion and Metabolic Compartmentation: Mitochondria and Chloroplasts	811
Chapter 15	Cell Signaling	873
Chapter 16	The Cytoskeleton	949
Chapter 17	The Cell Cycle	1027
Chapter 18	Cell Death	1089
PART V	CELLS IN THEIR SOCIAL CONTEXT	
Chapter 19	Cell Junctions and the Extracellular Matrix	1105
Chapter 20	Cancer	1163
Chapter 21	Development of Multicellular Organisms	1217
Chapter 22	Stem Cells in Tissue Homeostasis and Regeneration	1279
Chapter 23	Pathogens and Infection	1313
Chapter 24	The Innate and Adaptive Immune Systems	1353

Special Features

TABLE 1–2	Some Model Organisms and Their Genomes	29
TABLE 2–1	Covalent and Noncovalent Chemical Bonds	51
TABLE 2–2	Relationship Between the Standard Free-Energy Change, ΔG° , and the Equilibrium Constant	69
PANEL 2–1	Chemical Bonds and Groups Commonly Encountered in Biological Molecules	94
PANEL 2–2	Water and Its Influence on the Behavior of Biological Molecules	96
PANEL 2–3	The Principal Types of Weak Noncovalent Bonds That Hold Macromolecules Together	98
PANEL 2–4	An Outline of Some of the Types of Sugars Commonly Found in Cells	100
PANEL 2–5	Fatty Acids and Other Lipids	102
PANEL 2–6	A Survey of the Nucleotides	104
PANEL 2–7	Free Energy and Biological Reactions	106
PANEL 2–8	Details of the 10 Steps of Glycolysis	108
PANEL 2–9	The Complete Citric Acid Cycle	110
PANEL 3–1	The 20 Amino Acids Found in Proteins	118
TABLE 3–3	Macromolecular Machines Compared to Biomolecular Condensates and Membrane-enclosed Compartments	175
TABLE 3–4	Some Molecules Covalently Attached to Proteins That Regulate Protein Function	175
TABLE 4–1	Some Vital Statistics for the Human Genome	194
TABLE 5–4	Three Major Classes of Transposable Elements	308
TABLE 6–1	Principal Types of RNAs Produced in Cells	327
PANEL 7–1	Common Structural Motifs in Transcription Regulators	404
PANEL 8–1	Review of Classical Genetics	520
PANEL 9–1	Protein Structure Determination Using CryoEM	594
TABLE 11–1	A Comparison of Inorganic Ion Concentrations Inside and Outside a Typical Mammalian Cell	638
PANEL 11–1	The Derivation of the Nernst Equation	656
TABLE 12–1	Relative Volumes Occupied by the Major Intracellular Compartments in a Liver Cell (Hepatocyte)	684
TABLE 14–1	Quantity of Organelles and Organelle DNA in Some Cells and Tissues	815
TABLE 14–2	Mitochondrial Functions	819
PANEL 14–1	Redox Potentials	825
TABLE 14–3	Product Yields from the Oxidation of Sugars and Fats	836
TABLE 15–3	Four Major Families of Heterotrimeric G Proteins	907
TABLE 15–4	Some Extracellular Signal Proteins That Act Via RTKs	911
TABLE 15–5	The Ras Superfamily of Monomeric GTPases	915
TABLE 15–6	Some Extracellular Signal Proteins That Act Through Cytokine Receptors and the JAK–STAT Signaling Pathway	925
PANEL 16–2	The Polymerization of Actin and Tubulin	960
TABLE 16–1	Chemical Inhibitors of Actin and Microtubules	964
PANEL 16–3	Actin Filaments	965
PANEL 16–4	Microtubules	994
TABLE 16–2	Major Types of Intermediate Filament Proteins in Vertebrate Cells	1007
TABLE 17–1	The Major Cyclins and Cdks of Vertebrates and Budding Yeast	1034
TABLE 17–2	Summary of the Major Cell-Cycle Regulatory Proteins	1041
PANEL 17–1	The Principal Stages of M Phase (Mitosis and Cytokinesis) in an Animal Cell	1048
TABLE 19–1	Anchoring Junctions	1107
TABLE 19–2	Some Types of Collagen and Their Properties	1134
TABLE 19–3	Some Types of Integrins	1149
TABLE 20–2	Viruses Associated with Human Cancers	1202
TABLE 22–1	Blood Cells	1286
TABLE 23–1	Viruses That Cause Human Disease	1323
TABLE 24–2	Properties of the Major Classes of Antibodies in Humans	1375
TABLE 24–3	Properties of Human Class I and Class II MHC Proteins	1388
TABLE 24–4	Some Vaccines Approved for Human Use	1398

Contents

Chapter 1 Cells, Genomes, and the Diversity of Life	1		
THE UNIVERSAL FEATURES OF LIFE ON EARTH	2		
All Cells Store Their Hereditary Information in the Form of Double-Strand DNA Molecules	2	The Expression Levels of All the Genes of an Organism Can Be Determined	37
All Cells Replicate Their Hereditary Information by Templated Polymerization	3	<i>Arabidopsis</i> Has Been Chosen as a Model Plant	38
All Cells Transcribe Portions of Their DNA into RNA Molecules	5	The World of Animal Cells Is Mainly Represented by a Worm, a Fly, a Fish, a Mouse, and a Human	38
All Cells Use Proteins as Catalysts	6	Studies in the Fruit Fly <i>Drosophila</i> Provide a Key to Vertebrate Development	39
All Cells Translate RNA into Protein in the Same Way	6	The Frog and the Zebrafish Provide Highly Accessible Vertebrate Models	40
Each Protein Is Encoded by a Specific Gene	7	The Mouse Is the Predominant Mammalian Model Organism	41
Life Requires a Continual Input of Free Energy	7	The COVID-19 Pandemic Has Focused Scientists on the SARS-CoV-2 Coronavirus	42
All Cells Function as Biochemical Factories	8	Humans Are Unique in Reporting on Their Own Peculiarities To Understand Cells and Organisms Will Require Mathematics, Computers, and Quantitative Information	44
All Cells Are Enclosed in a Plasma Membrane Across Which Nutrients and Waste Materials Must Pass	8	Summary	45
Cells Operate at a Microscopic Scale Dominated by Random Thermal Motion	9	Problems	46
A Living Cell Can Exist with 500 Genes	10	References	47
Summary	10		
GENOME DIVERSIFICATION AND THE TREE OF LIFE	10	Chapter 2 Cell Chemistry and Bioenergetics	49
The Tree of Life Has Three Major Domains: Eukaryotes, Bacteria, and Archaea	11	THE CHEMICAL COMPONENTS OF A CELL	49
Eukaryotes Make Up the Domain of Life That Is Most Familiar to Us	13	Water Is Held Together by Hydrogen Bonds	50
On the Basis of Genome Analysis, Bacteria Are the Most Diverse Group of Organisms on the Planet	13	Four Types of Noncovalent Attractions Help Bring Molecules Together in Cells	51
Archaea: The Most Mysterious Domain of Life	15	Some Polar Molecules Form Acids and Bases in Water	52
Organisms Occupy Most of Our Planet	15	A Cell Is Formed from Carbon Compounds	53
Cells Can Be Powered by a Wide Variety of Free-Energy Sources	15	Cells Contain Four Major Families of Small Organic Molecules	53
Some Cells Fix Nitrogen and Carbon Dioxide for Other Cells	17	The Chemistry of Cells Is Dominated by Macromolecules with Remarkable Properties	54
Genomes Diversify Over Evolutionary Time, Producing New Types of Organisms	18	Noncovalent Bonds Specify Both the Precise Shape of a Macromolecule and Its Binding to Other Molecules	55
New Genes Are Generated from Preexisting Genes	19	Summary	56
Gene Duplications Give Rise to Families of Related Genes Within a Single Genome	20	CATALYSIS AND THE USE OF ENERGY BY CELLS	57
The Function of a Gene Can Often Be Deduced from Its Nucleotide Sequence	20	Cell Metabolism Is Organized by Enzymes	57
More Than 200 Gene Families Are Common to All Three Domains of Life	21	Biological Order Is Made Possible by the Release of Heat Energy from Cells	58
Summary	21	Cells Obtain Energy by the Oxidation of Organic Molecules	61
EUKARYOTES AND THE ORIGIN OF THE EUKARYOTIC CELL	22	Oxidation and Reduction Involve Electron Transfers	62
Eukaryotic Cells Contain a Variety of Organelles	23	Enzymes Lower the Activation-Energy Barriers That Block Chemical Reactions	63
Mitochondria Evolved from a Symbiotic Bacterium Captured by an Ancient Archaeon	25	Enzymes Can Drive Substrate Molecules Along Specific Reaction Pathways	64
Chloroplasts Evolved from a Symbiotic Photosynthetic Bacterium Engulfed by an Ancient Eukaryotic Cell	26	How Enzymes Find Their Substrates: The Enormous Rapidity of Molecular Motions	65
Eukaryotes Have Hybrid Genomes	27	The Free-Energy Change for a Reaction, ΔG , Determines Whether It Can Occur Spontaneously	66
Eukaryotic Genomes Are Big	28	The Concentration of Reactants Influences the Free-Energy Change and a Reaction's Direction	67
Eukaryotic Genomes Are Rich in Regulatory DNA	28	The Standard Free-Energy Change, ΔG° , Makes It Possible to Compare the Energetics of Different Reactions	67
Eukaryotic Genomes Define the Program of Multicellular Development	29	The Equilibrium Constant and ΔG° Are Readily Derived from Each Other	68
Many Eukaryotes Live as Solitary Cells	30	The Free-Energy Changes of Coupled Reactions Are Additive	69
Summary	31	Activated Carrier Molecules Are Essential for Biosynthesis	69
MODEL ORGANISMS	31	The Formation of an Activated Carrier Is Coupled to an Energetically Favorable Reaction	70
Mutations Reveal the Functions of Genes	32	ATP Is the Most Widely Used Activated Carrier Molecule	71
Molecular Biology Began with a Spotlight on One Bacterium and Its Viruses	33	Energy Stored in ATP Is Often Harnessed to Join Two Molecules Together	72
The Focus on <i>E. coli</i> as a Model Organism Has Accelerated Many Subsequent Discoveries	35		
A Yeast Serves as a Minimal Model Eukaryote	36		

Attractions Between Nucleosomes Compact the Chromatin Fiber	202	We Can Trace Human History by Analyzing Genomes	244
Summary	203	The Sequencing of Hundreds of Thousands of Human Genomes Reveals Much Variation	245
THE EFFECT OF CHROMATIN STRUCTURE ON DNA FUNCTION	203	Most of the Variants Observed in the Human Population Are Common Alleles, with at Most a Weak Effect on Phenotype	246
Different Regions of the Human Genome Are Packaged Very Differently in Chromatin	204	Forensic Analyses Exploit Special DNA Sequences with Unusually High Mutation Rates	247
Heterochromatin Is Highly Condensed and Restricts Gene Expression	204	An Understanding of Human Variation Is Critical for Improving Medicine	248
The Heterochromatic State Can Spread Along a Chromosome and Be Inherited from One Cell Generation to the Next	205	Summary	248
The Core Histones Are Covalently Modified at Many Different Sites	206	Problems	249
Chromatin Acquires Additional Variety Through the Site-specific Insertion of a Small Set of Histone Variants	208	References	251
Covalent Modifications and Histone Variants Can Act in Concert to Control Chromosome Functions	208	Chapter 5 DNA Replication, Repair, and Recombination	253
A Complex of Reader and Writer Proteins Can Spread Specific Chromatin Modifications Along a Chromosome	210	THE MAINTENANCE OF DNA SEQUENCES	253
Barrier DNA-Protein Complexes Block the Spread of Reader-Writer Complexes and Thereby Separate Neighboring Chromatin Domains	212	Mutation Rates Are Extremely Low	253
Centromeres Have a Special, Inherited Chromatin Structure	213	Low Mutation Rates Are Necessary for Life as We Know It	254
Some Forms of Chromatin Can Be Directly Inherited	215	Summary	255
The Abnormal Perturbations of Heterochromatin That Arise During Tumor Progression Contribute to Many Cancers	215	DNA REPLICATION MECHANISMS	255
Summary	217	Base-pairing Underlies DNA Replication and DNA Repair	255
THE GLOBAL STRUCTURE OF CHROMOSOMES	217	The DNA Replication Fork Is Asymmetrical	256
Chromosomes Are Folded into Large Loops of Chromatin	217	The High Fidelity of DNA Replication Requires Several Proofreading Mechanisms	258
Polytene Chromosomes Are Uniquely Useful for Visualizing Chromatin Structures	218	DNA Replication in the 5'-to-3' Direction Allows Efficient Error Correction	260
Chromosome Loops Decondense When the Genes Within Them Are Expressed	220	A Special Nucleotide-polymerizing Enzyme Synthesizes Short RNA Primer Molecules	260
Mammalian Interphase Chromosomes Occupy Discrete Territories in the Nucleus, with Their Heterochromatin and Euchromatin Distributed Differently	220	Special Proteins Help to Open Up the DNA Double Helix in Front of the Replication Fork	261
A Biochemical Technique Called Hi-C Reveals Details of Chromosome Organization	221	A Sliding Ring Holds a Moving DNA Polymerase onto the DNA	262
Chromosomal DNA Is Organized into Loops by Large Protein Rings	223	The Proteins at a Replication Fork Cooperate to Form a Replication Machine	263
Euchromatin and Heterochromatin Separate Spatially in the Nucleus	225	DNA Replication Is Fundamentally Similar in Eukaryotes and Bacteria	265
Mitotic Chromosomes Are Highly Condensed	227	A Strand-directed Mismatch Repair System Removes Replication Errors That Remain in the Wake of the Replication Machine	267
Summary	228	The Accidental Incorporation of Ribonucleotides During DNA Replication Is Corrected	269
HOW GENOMES EVOLVE	229	DNA Topoisomerases Prevent DNA Tangling During Replication	269
Genome Comparisons Reveal Functional DNA Sequences by Their Conservation Throughout Evolution	230	Summary	272
Genome Alterations Are Caused by Failures of the Normal Mechanisms for Copying and Maintaining DNA, as Well as by Transposable DNA Elements	231	THE INITIATION AND COMPLETION OF DNA REPLICATION IN CHROMOSOMES	272
The Genome Sequences of Two Species Differ in Proportion to the Length of Time Since They Have Separately Evolved	232	DNA Synthesis Begins at Replication Origins	272
Phylogenetic Trees Constructed from a Comparison of DNA Sequences Trace the Relationships of All Organisms	233	Bacterial Chromosomes Typically Have a Single Origin of DNA Replication	273
A Comparison of Human and Mouse Chromosomes Shows How the Structures of Genomes Diverge	234	Eukaryotic Chromosomes Contain Multiple Origins of Replication	273
The Size of a Vertebrate Genome Reflects the Relative Rates of DNA Addition and DNA Loss in a Lineage	236	In Eukaryotes, DNA Replication Takes Place During Only One Part of the Cell Cycle	276
Multispecies Sequence Comparisons Identify Many Conserved DNA Sequences of Unknown Function	237	Eukaryotic Origins of Replication Are "Licensed" for Replication by the Assembly of an Origin Recognition Complex	276
Changes in Previously Conserved Sequences Can Help Decipher Critical Steps in Evolution	238	Features of the Human Genome That Specify Origins of Replication Remain to Be Fully Understood	277
Mutations in the DNA Sequences That Control Gene Expression Have Driven Many of the Evolutionary Changes in Vertebrates	239	Properties of the ORC Ensure That Each Region of the DNA Is Replicated Once and Only Once in Each S Phase	277
Gene Duplication Also Provides an Important Source of Genetic Novelty During Evolution	240	New Nucleosomes Are Assembled Behind the Replication Fork	279
Duplicated Genes Diverge	240	Termination of DNA Replication Occurs Through the Ordered Disassembly of the Replication Fork	280
The Evolution of the Globin Gene Family Shows How DNA Duplications Contribute to the Evolution of Organisms	241	Telomerase Replicates the Ends of Chromosomes	281
Genes Encoding New Proteins Can Be Created by the Recombination of Exons	242	Telomeres Are Packaged into Specialized Structures That Protect the Ends of Chromosomes	282
Neutral Mutations Often Spread to Become Fixed in a Population, with a Probability That Depends on Population Size	243	Telomere Length Is Regulated by Cells and Organisms	282
		Summary	284
		DNA REPAIR	284
		Without DNA Repair, Spontaneous DNA Damage Would Rapidly Change DNA Sequences	286
		The DNA Double Helix Is Readily Repaired	288
		DNA Damage Can Be Removed by More Than One Pathway	288

Coupling Nucleotide Excision Repair to Transcription Ensures That the Cell's Most Important DNA Is Efficiently Repaired	290	In Eukaryotes, Transcription Initiation Also Requires Activator, Mediator, and Chromatin-modifying Proteins	334
The Chemistry of the DNA Bases Facilitates Damage Detection	290	Transcription Elongation in Eukaryotes Requires Accessory Proteins	335
Special Translesion DNA Polymerases Are Used in Emergencies	292	Transcription Creates Superhelical Tension	335
Double-Strand Breaks Are Efficiently Repaired	292	Transcription Elongation in Eukaryotes Is Tightly Coupled to RNA Processing	337
DNA Damage Delays Progression of the Cell Cycle	295	RNA Capping Is the First Modification of Eukaryotic Pre-mRNAs	338
Summary	295	RNA Splicing Removes Intron Sequences from Newly Transcribed Pre-mRNAs	339
HOMOLOGOUS RECOMBINATION	296	Nucleotide Sequences Signal Where Splicing Occurs	341
Homologous Recombination Has Common Features in All Cells	296	RNA Splicing Is Performed by the Spliceosome	341
DNA Base-pairing Guides Homologous Recombination	296	The Spliceosome Uses ATP Hydrolysis to Produce a Complex Series of RNA–RNA Rearrangements	343
Homologous Recombination Can Flawlessly Repair Double-Strand Breaks in DNA	297	Other Properties of Pre-mRNA and Its Synthesis Help to Explain the Choice of Proper Splice Sites	345
Specialized Processing of Double-Strand Breaks Commits Repair to Homologous Recombination	298	RNA Splicing Has Remarkable Plasticity	346
Strand Exchange Is Directed by the RecA/Rad51 Protein	298	Spliceosome-catalyzed RNA Splicing Evolved from RNA Self-splicing Mechanisms	347
Homologous Recombination Can Rescue Broken and Stalled DNA Replication Forks	299	RNA-processing Enzymes Generate the 3' End of Eukaryotic mRNAs	348
DNA Repair by Homologous Recombination Entails Risks to the Cell	300	Mature Eukaryotic mRNAs Are Selectively Exported from the Nucleus	349
Homologous Recombination Is Crucial for Meiosis	301	Noncoding RNAs Are Also Synthesized and Processed in the Nucleus	351
Meiotic Recombination Begins with a Programmed Double-Strand Break	302	The Nucleolus Is a Ribosome-producing Factory	353
Holliday Junctions Are Recognized by Enzymes That Drive Branch Migration	302	The Nucleus Contains a Variety of Subnuclear Biomolecular Condensates	355
Homologous Recombination Produces Crossovers Between Maternal and Paternal Chromosomes During Meiosis	304	Summary	357
Homologous Recombination Often Results in Gene Conversion	305	FROM RNA TO PROTEIN	358
Summary	306	An mRNA Sequence Is Decoded in Sets of Three Nucleotides	358
TRANSPPOSITION AND CONSERVATIVE SITE-SPECIFIC RECOMBINATION	306	tRNA Molecules Match Amino Acids to Codons in mRNA	359
Through Transposition, Mobile Genetic Elements Can Insert into Any DNA Sequence	307	tRNAs Are Covalently Modified Before They Exit from the Nucleus	361
DNA-only Transposons Can Move by a Cut-and-Paste Mechanism	307	Specific Enzymes Couple Each Amino Acid to Its Appropriate tRNA Molecule	361
Some DNA-only Transposons Move by Replicating Themselves	309	Editing by tRNA Synthetases Ensures Accuracy	363
Some Viruses Use a Transposition Mechanism to Move Themselves into Host-Cell Chromosomes	309	Amino Acids Are Added to the C-terminal End of a Growing Polypeptide Chain	364
Some RNA Viruses Replicate and Express Their Genomes Without Using DNA as an Intermediate	311	The RNA Message Is Decoded in Ribosomes	365
Retroviral-like Retrotransposons Resemble Retroviruses, but Cannot Move from Cell to Cell	313	Elongation Factors Drive Translation Forward and Improve Its Accuracy	368
A Large Fraction of the Human Genome Is Composed of Nonretroviral Retrotransposons	313	Induced Fit and Kinetic Proofreading Help Biological Processes Overcome the Inherent Limitations of Complementary Base-Pairing	369
Different Transposable Elements Predominate in Different Organisms	314	Accuracy in Translation Requires a Large Expenditure of Free Energy	370
Genome Sequences Reveal the Approximate Times at Which Transposable Elements Have Moved	314	The Ribosome Is a Ribozyme	371
Conservative Site-specific Recombination Can Reversibly Rearrange DNA	315	Nucleotide Sequences in mRNA Signal Where to Start Protein Synthesis	373
Conservative Site-specific Recombination Can Be Used to Turn Genes On or Off	316	Stop Codons Mark the End of Translation	374
Bacterial Conservative Site-specific Recombinases Have Become Powerful Tools for Cell and Developmental Biologists	317	Proteins Are Made on Polyribosomes	375
Summary	317	There Are Minor Variations in the Standard Genetic Code	375
Problems	318	Inhibitors of Prokaryotic Protein Synthesis Are Useful as Antibiotics	376
References	320	Quality-Control Mechanisms Act to Prevent Translation of Damaged mRNAs	378
		Stalled Ribosomes Can Be Rescued	379
		The Ribosome Coordinates the Folding, Enzymatic Modification, and Assembly of Newly Synthesized Proteins	380
Chapter 6 How Cells Read the Genome: From DNA to Protein	321	Molecular Chaperones Help Guide the Folding of Most Proteins	380
FROM DNA TO RNA	323	Proper Folding of Newly Synthesized Proteins Is Also Aided by Translation Speed and Subunit Assembly	383
RNA Molecules Are Single-Stranded	324	Proteins That Ultimately Fail to Fold Correctly Are Marked for Destruction by Polyubiquitin	384
Transcription Produces RNA Complementary to One Strand of DNA	325	The Proteasome Is a Compartmentalized Protease with Sequestered Active Sites	384
RNA Polymerases Carry Out DNA Transcription	325	Many Proteins Are Controlled by Regulated Destruction	386
Cells Produce Different Categories of RNA Molecules	327	There Are Many Steps from DNA to Protein	387
Signals Encoded in DNA Tell RNA Polymerase Where to Start and Stop	328	Summary	388
Bacterial Transcription Start and Stop Signals Are Heterogeneous in Nucleotide Sequence	329	THE RNA WORLD AND THE ORIGINS OF LIFE	389
Transcription Initiation in Eukaryotes Requires Many Proteins	331	Single-Strand RNA Molecules Can Fold into Highly Elaborate Structures	390
To Initiate Transcription, RNA Polymerase II Requires a Set of General Transcription Factors	332	Ribozymes Can Be Produced in the Laboratory	390

RNA Can Both Store Information and Catalyze Chemical Reactions	391	Transcription Circuits Allow the Cell to Carry Out Logic Operations	433
How Did Protein Synthesis Evolve?	392	Summary	434
All Present-Day Cells Use DNA as Their Hereditary Material	393	MECHANISMS THAT REINFORCE CELL MEMORY IN PLANTS AND ANIMALS	435
Summary	393	Patterns of DNA Methylation Can Be Inherited When Vertebrate Cells Divide	435
Problems	394	CG-Rich Islands Are Associated with Many Genes in Mammals	436
References	395	Genomic Imprinting Is Based on DNA Methylation	438
		A Chromosome-wide Alteration in Chromatin Structure Can Be Inherited	440
Chapter 7 Control of Gene Expression	397	The Mammalian X-Inactivation in Females Is Triggered by the Synthesis of a Long Noncoding RNA	442
AN OVERVIEW OF GENE CONTROL	397	Stable Patterns of Gene Expression Can Be Transmitted to Daughter Cells	443
The Different Cell Types of a Multicellular Organism Contain the Same DNA	397	Summary	445
Different Cell Types Synthesize Different Sets of RNAs and Proteins	398	POST-TRANSCRIPTIONAL CONTROLS	445
The Spectrum of mRNAs Present in a Cell Can Be Used to Accurately Identify the Cell Type	400	Transcription Attenuation Causes the Premature Termination of Some RNA Molecules	445
External Signals Can Cause a Cell to Change the Expression of Its Genes	400	Riboswitches Probably Represent Ancient Forms of Gene Control	446
Gene Expression Can Be Regulated at Many of the Steps in the Pathway from DNA to RNA to Protein	401	Alternative RNA Splicing Can Produce Different Forms of a Protein from the Same Gene	446
Summary	402	The Definition of a Gene Has Been Modified Since the Discovery of Alternative RNA Splicing	448
CONTROL OF TRANSCRIPTION BY SEQUENCE-SPECIFIC DNA-BINDING PROTEINS	402	Back Splicing Can Produce Circular RNA Molecules	449
The Sequence of Nucleotides in the DNA Double Helix Can Be Read by Proteins	402	A Change in the Site of RNA Transcript Cleavage and Poly-A Addition Can Change the C-terminus of a Protein	449
Transcription Regulators Contain Structural Motifs That Can Read DNA Sequences	403	Nucleotides in mRNA Can Be Covalently Modified	450
Dimerization of Transcription Regulators Increases Their Affinity and Specificity for DNA	406	RNA Editing Can Change the Meaning of the RNA Message	451
Many Transcription Regulators Bind Cooperatively to DNA	407	The Human AIDS Virus Illustrates How RNA Transport from the Nucleus Can Be Regulated	452
Nucleosome Structure Promotes Cooperative Binding of Transcription Regulators	408	mRNAs Can Be Localized to Specific Regions of the Cytosol	453
DNA-Binding by Transcription Regulators Is Dynamic	409	Untranslated Regions of mRNAs Control Their Translation	456
Summary	410	The Phosphorylation of an Initiation Factor Regulates Protein Synthesis Globally	457
TRANSCRIPTION REGULATORS SWITCH GENES ON AND OFF	410	Initiation at AUG Codons Upstream of the Translation Start Can Regulate Eukaryotic Translation Initiation	458
The Tryptophan Repressor Switches Genes Off	410	Internal Ribosome Entry Sites Also Provide Opportunities for Translational Control	458
Repressors Turn Genes Off and Activators Turn Them On	411	Changes in mRNA Stability Can Control Gene Expression	459
Both an Activator and a Repressor Control the <i>Lac</i> Operon	412	Regulation of mRNA Stability Involves P-bodies and Stress Granules	461
DNA Looping Can Occur During Bacterial Gene Regulation	412	Summary	462
Complex Switches Control Gene Transcription in Eukaryotes	414	REGULATION OF GENE EXPRESSION BY NONCODING RNAs	462
A Eukaryotic Gene Control Region Includes Many <i>cis</i> -Regulatory Sequences	414	Small Noncoding RNA Transcripts Regulate Many Animal and Plant Genes Through RNA Interference	462
Eukaryotic Transcription Regulators Work in Groups	415	miRNAs Regulate mRNA Translation and Stability	463
Activator Proteins Promote the Assembly of RNA Polymerase at the Start Point of Transcription	416	RNA Interference Also Serves as a Cell Defense Mechanism	464
Eukaryotic Transcription Activators Direct the Modification of Local Chromatin Structure	417	RNA Interference Can Direct Heterochromatin Formation	465
Some Transcription Activators Work by Releasing Paused RNA Polymerase	418	piRNAs Protect the Germ Line from Transposable Elements	466
Transcription Activators Work Synergistically	419	RNA Interference Has Become a Powerful Experimental Tool	467
Condensate Formation Likely Increases the Efficiency of Transcription Initiation	420	Cells Have Additional Mechanisms to Hold Transposons and Integrated Viral Genomes in Check	467
Eukaryotic Transcription Repressors Can Inhibit Transcription in Several Ways	420	Bacteria Use Small Noncoding RNAs to Protect Themselves from Viruses	468
Insulator DNA Sequences Prevent Eukaryotic Transcription	422	Long Noncoding RNAs Have Diverse Functions in the Cell	469
Regulators from Influencing Distant Genes	422	Summary	471
Summary	422	Problems	472
MOLECULAR GENETIC MECHANISMS THAT CREATE AND MAINTAIN SPECIALIZED CELL TYPES	423	References	474
Complex Genetic Switches That Regulate <i>Drosophila</i> Development Are Built Up from Smaller Modules	423	Chapter 8 Analyzing Cells, Molecules, and Systems	475
The <i>Drosophila Eve</i> Gene Is Regulated by Combinatorial Controls	424	ISOLATING CELLS AND GROWING THEM IN CULTURE	476
Transcription Regulators Are Brought into Play by Extracellular Signals	426	Cells Can Be Isolated from Tissues and Grown in Culture	476
Combinatorial Gene Control Creates Many Different Cell Types	427	Eukaryotic Cell Lines Are a Widely Used Source of Homogeneous Cells	478
Specialized Cell Types Can Be Experimentally Reprogrammed to Become Pluripotent Stem Cells	428	Hybridoma Cell Lines Are Factories That Produce Monoclonal Antibodies	478
Combinations of Master Transcription Regulators Specify Cell Types by Controlling the Expression of Many Genes	429	Summary	480
Specialized Cells Must Rapidly Turn Some Genes On and Off	430	PURIFYING PROTEINS	480
Differentiated Cells Maintain Their Identity	431	Cells Can Be Separated into Their Component Fractions	480
		Cell Extracts Provide Accessible Systems to Study Cell Functions	482
		Proteins Can Be Separated by Chromatography	483

Immunoprecipitation Is a Rapid Affinity Purification Method	486	Expression of Individual Genes Can Be Measured Using	536
Genetically Engineered Tags Provide an Easy Way to Purify Proteins	486	Quantitative RT-PCR	
Purified Cell-free Systems Are Required for the Precise Dissection of Molecular Functions	486	Global Analysis of mRNAs by RNA-seq Provides a Snapshot of Gene Expression	536
Summary	487	Genome-wide Chromatin Immunoprecipitation Identifies Sites on the Genome Occupied by Transcription Regulators	538
ANALYZING PROTEINS	487	Ribosome Profiling Reveals Which mRNAs Are Being Translated in the Cell	538
Proteins Can Be Separated by SDS Polyacrylamide-Gel Electrophoresis	487	Recombinant DNA Methods Have Revolutionized Human Health	539
Two-dimensional Gel Electrophoresis Provides Greater Protein Separation	489	Transgenic Plants Are Important for Agriculture	540
Specific Proteins Can Be Detected by Blotting with Antibodies	490	Summary	542
Hydrodynamic Measurements Reveal the Size and Shape of a Protein Complex	490	MATHEMATICAL ANALYSIS OF CELL FUNCTION	542
Mass Spectrometry Provides a Highly Sensitive Method for Identifying Unknown Proteins	491	Regulatory Networks Depend on Molecular Interactions	543
Sets of Interacting Proteins Can Be Identified by Biochemical Methods	493	Differential Equations Help Us Predict Transient Behavior	545
Optical Methods Can Monitor Protein Interactions	493	Promoter Activity and Protein Degradation Affect the Rate of Change of Protein Concentration	546
Protein Structure Can Be Determined Using X-ray Diffraction	494	The Time Required to Reach Steady State Depends on Protein Lifetime	547
NMR Can Be Used to Determine Protein Structure in Solution	496	Quantitative Methods Are Similar for Transcription Repressors and Activators	548
Protein Sequence and Structure Provide Clues About Protein Function	497	Negative Feedback Is a Powerful Strategy in Cell Regulation	549
Summary	498	Delayed Negative Feedback Can Induce Oscillations	549
ANALYZING AND MANIPULATING DNA	498	DNA Binding by a Repressor or an Activator Can Be Cooperative	551
Restriction Nucleases Cut Large DNA Molecules into Specific Fragments	498	Positive Feedback Is Important for Switchlike Responses and Bistability	551
Gel Electrophoresis Separates DNA Molecules of Different Sizes	499	Robustness Is an Important Characteristic of Biological Networks	553
Purified DNA Molecules Can Be Specifically Labeled with Radioisotopes or Chemical Markers <i>in Vitro</i>	501	Two Transcription Regulators That Bind to the Same Gene Promoter Can Exert Combinatorial Control	554
Genes Can Be Cloned Using Bacteria	501	An Incoherent Feed-forward Interaction Generates Pulses	555
An Entire Genome Can Be Represented in a DNA Library	503	A Coherent Feed-forward Interaction Detects Persistent Inputs	556
Hybridization Provides a Powerful but Simple Way to Detect Specific Nucleotide Sequences	505	The Same Network Can Behave Differently in Different Cells Because of Stochastic Effects	557
Genes Can Be Cloned <i>in Vitro</i> Using PCR	506	Several Computational Approaches Can Be Used to Model the Reactions in Cells	557
PCR Is Also Used for Diagnostic and Forensic Applications	507	Statistical Methods Are Critical for the Analysis of Biological Data	558
PCR and Synthetic DNA Are Ideal Sources of Specific Gene Sequences for Cloning	510	Summary	558
DNA Cloning Allows Any Protein to Be Produced in Large Amounts	511	Problems	559
DNA Can Be Sequenced Rapidly by Dideoxy Sequencing	512	References	561
Next-Generation Sequencing Methods Have Revolutionized DNA and RNA Analysis	514		
To Be Useful, Genome Sequences Must Be Annotated	516	Chapter 9 Visualizing Cells and Their Molecules	563
Summary	518		
STUDYING GENE FUNCTION AND EXPRESSION	518	LOOKING AT CELLS AND MOLECULES IN THE LIGHT MICROSCOPE	563
Classical Genetic Screens Identify Random Mutants with Specific Abnormalities	519	The Conventional Light Microscope Can Resolve Details 0.2 μm Apart	564
Mutations Can Cause Loss or Gain of Protein Function	522	Photon Noise Creates Additional Limits to Resolution When Light Levels Are Low	567
Complementation Tests Reveal Whether Two Mutations Are in the Same Gene or Different Genes	523	Living Cells Are Seen Clearly in a Phase-Contrast or a Differential-Interference-Contrast Microscope	567
Gene Products Can Be Ordered in Pathways by Epistasis Analysis	523	Images Can Be Enhanced and Analyzed by Digital Techniques	568
Mutations Responsible for a Phenotype Can Be Identified Through DNA Analysis	524	Intact Tissues Are Usually Fixed and Sectioned Before Microscopy	569
Rapid and Cheap DNA Sequencing Has Revolutionized Human Genetic Studies	524	Specific Molecules Can Be Located in Cells by Fluorescence Microscopy	570
Linked Blocks of Polymorphisms Have Been Passed Down from Our Ancestors	525	Antibodies Can Be Used to Detect Specific Proteins	572
Sequence Variants Can Aid the Search for Mutations Associated with Disease	526	Individual Proteins Can Be Fluorescently Tagged in Living Cells and Organisms	573
Genomics Is Accelerating the Discovery of Rare Mutations That Predispose Us to Serious Disease	527	Protein Dynamics Can Be Followed in Living Cells	575
The Cellular Functions of a Known Gene Can Be Studied with Genome Engineering	527	Fluorescent Biosensors Can Monitor Cell Signaling	576
Animals and Plants Can Be Genetically Altered	528	Imaging of Complex Three-dimensional Objects Is Possible with the Optical Microscope	577
The Bacterial CRISPR System Has Been Adapted to Edit Genomes in a Wide Variety of Species	530	The Confocal Microscope Produces Optical Sections by Excluding Out-of-Focus Light	578
Large Collections of Engineered Mutations Provide a Tool for Examining the Function of Every Gene in an Organism	531	Superresolution Fluorescence Techniques Can Overcome Diffraction-limited Resolution	580
RNA Interference Is a Simple and Rapid Way to Test Gene Function	533	Single-Molecule Localization Microscopy Also Delivers Superresolution	583
Reporter Genes Reveal When and Where a Gene Is Expressed	534	Expanding the Specimen Can Offer Higher Resolution, but with a Conventional Microscope	585
<i>In Situ</i> Hybridization Can Reveal the Location of mRNAs and Noncoding RNAs	535	Large Multicellular Structures Can Be Imaged Over Time	586
		Single Molecules Can Be Visualized by Total Internal Reflection Fluorescence Microscopy	587
		Summary	588

LOOKING AT CELLS AND MOLECULES IN THE ELECTRON MICROSCOPE

The Electron Microscope Resolves the Fine Structure of the Cell	588
Biological Specimens Require Special Preparation for Electron Microscopy	589
Heavy Metals Can Provide Additional Contrast	590
Images of Surfaces Can Be Obtained by Scanning Electron Microscopy	591
Electron Microscope Tomography Allows the Molecular Architecture of Cells to Be Seen in Three Dimensions	593
Cryo-electron Microscopy Can Determine Molecular Structures at Atomic Resolution	595
Light Microscopy and Electron Microscopy Are Mutually Beneficial	597
Using Microscopy to Study Cells Always Involves Trade-Offs	598
Summary	599
Problems	600
References	601

Chapter 10 Membrane Structure 603**THE LIPID BILAYER 604**

Glycerophospholipids, Sphingolipids, and Sterols Are the Major Lipids in Cell Membranes	605
Phospholipids Spontaneously Form Bilayers	606
The Lipid Bilayer Is a Two-dimensional Fluid	608
The Fluidity of a Lipid Bilayer Depends on Its Composition	609
Despite Their Fluidity, Lipid Bilayers Can Form Domains of Different Compositions	610
Lipid Droplets Are Surrounded by a Phospholipid Monolayer	611
The Asymmetry of the Lipid Bilayer Is Functionally Important	612
Glycolipids Are Found on the Surface of All Eukaryotic Plasma Membranes	613
Summary	614
MEMBRANE PROTEINS 615	
Membrane Proteins Can Be Associated with the Lipid Bilayer in Various Ways	615
Lipid Anchors Control the Membrane Localization of Some Signaling Proteins	616
In Most Transmembrane Proteins, the Polypeptide Chain Crosses the Lipid Bilayer in an α -Helical Conformation	617
Transmembrane α Helices Often Interact with One Another	619
Some β Barrels Form Large Channels	619
Many Membrane Proteins Are Glycosylated	621
Membrane Proteins Can Be Solubilized and Purified in Detergents	622
Bacteriorhodopsin Is a Light-driven Proton (H^+) Pump That Traverses the Lipid Bilayer as Seven α Helices	625
Membrane Proteins Often Function as Large Complexes	627
Many Membrane Proteins Diffuse in the Plane of the Membrane	627
Cells Can Confine Proteins and Lipids to Specific Domains Within a Membrane	629
The Cortical Cytoskeleton Gives Membranes Mechanical Strength and Restricts Membrane Protein Diffusion	630
Membrane-bending Proteins Deform Bilayers	632
Summary	633
Problems	634
References	635

Chapter 11 Small-Molecule Transport and Electrical Properties of Membranes 637**PRINCIPLES OF MEMBRANE TRANSPORT 637**

Protein-free Lipid Bilayers Are Impermeable to Ions	638
There Are Two Main Classes of Membrane Transport Proteins: Transporters and Channels	638
Active Transport Is Mediated by Transporters Coupled to an Energy Source	639
Summary	640
TRANSPORTERS AND ACTIVE MEMBRANE TRANSPORT 640	
Active Transport Can Be Driven by Ion-Concentration Gradients	642
Transporters in the Plasma Membrane Regulate Cytosolic pH	644
An Asymmetric Distribution of Transporters in Epithelial Cells Underlies the Transcellular Transport of Solutes	645
There Are Three Classes of ATP-driven Pumps	646

A P-type ATPase Pumps Ca^{2+} into the Sarcoplasmic Reticulum in Muscle Cells	647
The Plasma Membrane Na^+-K^+ Pump Establishes Na^+ and K^+ Gradients Across the Plasma Membrane	648
ABC Transporters Constitute the Largest Family of Membrane Transport Proteins	649
Summary	651
CHANNELS AND THE ELECTRICAL PROPERTIES OF MEMBRANES 651	
Aquaporins Are Permeable to Water but Impermeable to Ions	652
Ion Channels Are Ion-selective and Fluctuate Between Open and Closed States	653
The Membrane Potential in Animal Cells Depends Mainly on K^+ Leak Channels and the K^+ Gradient Across the Plasma Membrane	655
The Resting Potential Decays Only Slowly When the Na^+-K^+ Pump Is Stopped	655
The Three-dimensional Structure of a Bacterial K^+ Channel Shows How an Ion Channel Can Work	657
Mechanosensitive Channels Allow Cells to Sense Their Physical Environment	659
The Function of a Neuron Depends on Its Elongated Structure	661
Voltage-gated Cation Channels Generate Action Potentials in Electrically Excitable Cells	662
Myelination Increases the Speed and Efficiency of Action Potential Propagation in Nerve Cells	666
Patch-Clamp Recording Indicates That Individual Ion Channels Open in an All-or-Nothing Fashion	666
Voltage-gated Cation Channels Are Evolutionarily and Structurally Related	668
Different Neuron Types Display Characteristic Stable Firing Properties	668
Transmitter-gated Ion Channels Convert Chemical Signals into Electrical Ones at Chemical Synapses	669
Chemical Synapses Can Be Excitatory or Inhibitory	670
The Acetylcholine Receptors at the Neuromuscular Junction Are Excitatory Transmitter-gated Cation Channels	671
Neurons Contain Many Types of Transmitter-gated Channels	672
Many Psychoactive Drugs Act at Synapses	673
Neuromuscular Transmission Involves the Sequential Activation of Five Different Sets of Ion Channels	673
Single Neurons Are Complex Computation Devices	674
Neuronal Computation Requires a Combination of at Least Three Kinds of K^+ Channels	675
Long-term Potentiation in the Mammalian Hippocampus Depends on Ca^{2+} Entry Through NMDA-Receptor Channels	677
The Use of Channelrhodopsins Has Revolutionized the Study of Neural Circuits	678
Summary	679
Problems	680
References	681

Chapter 12 Intracellular Organization and Protein Sorting 683**THE COMPARTMENTALIZATION OF CELLS 683**

All Eukaryotic Cells Have the Same Basic Set of Membrane-enclosed Organelles	683
Evolutionary Origins Explain the Topological Relationships of Organelles	686
Macromolecules Can Be Segregated Without a Surrounding Membrane	688
Multivalent Interactions Mediate Formation of Biomolecular Condensates	690
Biomolecular Condensates Create Biochemical Factories	690
Biomolecular Condensates Form and Disassemble in Response to Need	693
Proteins Can Move Between Compartments in Different Ways	694
Sorting Signals and Sorting Receptors Direct Proteins to the Correct Cell Address	695
Construction of Most Organelles Requires Information in the Organelle Itself	697
Summary	697

THE ENDOPLASMIC RETICULUM	698	The Assembly of a Clathrin Coat Drives Vesicle Formation	752
The ER Is Structurally and Functionally Diverse	698	Adaptor Proteins Select Cargo into Clathrin-coated Vesicles	753
Signal Sequences Were First Discovered in Proteins Imported into the Rough ER	701	Phosphoinositides Mark Organelles and Membrane Domains	754
A Signal-Recognition Particle (SRP) Directs the ER Signal Sequence to a Specific Receptor at the ER	702	Membrane-bending Proteins Help Deform the Membrane During Vesicle Formation	755
The Polypeptide Chain Passes Through a Signal Sequence-gated Aqueous Channel in the Translocator	705	Cytoplasmic Proteins Regulate the Pinching off and Uncoating of Coated Vesicles	756
Translocation Across the ER Membrane Does Not Always Require Ongoing Polypeptide Chain Elongation	707	Monomeric GTPases Control Coat Assembly	756
Transmembrane Proteins Contain Hydrophobic Segments That Are Recognized Like Signal Sequences	709	Coat-recruitment GTPases Participate in Coat Disassembly	758
Hydrophobic Segments of Multipass Transmembrane Proteins Are Interpreted Contextually to Determine Their Orientation	710	The Shape and Size of Transport Vesicles Are Diverse	759
Some Proteins Are Integrated into the ER Membrane by a Post-translational Mechanism	711	Rab Proteins Guide Transport Vesicles to Their Target Membrane	760
Some Membrane Proteins Acquire a Covalently Attached Glycosylphosphatidylinositol (GPI) Anchor	712	Rab Proteins Create and Change the Identity of an Organelle	761
Translocated Polypeptide Chains Fold and Assemble in the Lumen of the Rough ER	712	SNAREs Mediate Membrane Fusion	762
Most Proteins Synthesized in the Rough ER Are Glycosylated by the Addition of a Common <i>N</i> -Linked Oligosaccharide	714	Interacting SNAREs Need to Be Pried Apart Before They Can Function Again	763
Oligosaccharides Are Used as Tags to Mark the State of Protein Folding	715	Viruses Encode Specialized Membrane Fusion Proteins Needed for Cell Entry	764
Improperly Folded Proteins Are Exported from the ER and Degraded in the Cytosol	716	Summary	764
Misfolded Proteins in the ER Activate an Unfolded Protein Response	717	TRANSPORT FROM THE ENDOPLASMIC RETICULUM THROUGH THE GOLGI APPARATUS	765
The ER Assembles Most Lipid Bilayers	720	Proteins Leave the ER in COPII-coated Transport Vesicles	765
Membrane Contact Sites Between the ER and Other Organelles Facilitate Selective Lipid Transfer	722	Only Proteins That Are Properly Folded and Assembled Can Leave the ER	766
Summary	723	Vesicular Tubular Clusters Mediate Transport from the ER to the Golgi Apparatus	766
PEROXISOMES	723	The Retrieval Pathway to the ER Uses Sorting Signals	768
Peroxisomes Use Molecular Oxygen and Hydrogen Peroxide to Perform Oxidation Reactions	724	Many Proteins Are Selectively Retained in the Compartments in Which They Function	768
Short Signal Sequences Direct the Import of Proteins into Peroxisomes	724	The Golgi Apparatus Consists of an Ordered Series of Compartments	769
Summary	726	Oligosaccharide Chains Are Processed in the Golgi Apparatus	771
THE TRANSPORT OF PROTEINS INTO MITOCHONDRIA AND CHLOROPLASTS	726	Proteoglycans Are Assembled in the Golgi Apparatus	772
Translocation into Mitochondria Depends on Signal Sequences and Protein Translocators	727	What Is the Purpose of Glycosylation?	773
Mitochondrial Proteins Are Imported Post-translationally as Unfolded Polypeptide Chains	728	Transport Through the Golgi Apparatus Occurs by Multiple Mechanisms	774
Protein Import Is Powered by ATP Hydrolysis, a Membrane Potential, and Redox Potential	730	Golgi Matrix Proteins Help Organize the Stack	775
Transport into the Inner Mitochondrial Membrane Occurs Via Several Routes	731	Summary	776
Bacteria and Mitochondria Use Similar Mechanisms to Insert β Barrels into Their Outer Membrane	733	TRANSPORT FROM THE <i>TRANS</i> GOLGI NETWORK TO THE CELL EXTERIOR AND ENDOSOMES	776
Two Signal Sequences Direct Proteins to the Thylakoid Membrane in Chloroplasts	733	Many Proteins and Lipids Are Carried Automatically from the <i>Trans</i> Golgi Network to the Cell Surface	777
Summary	735	A Mannose 6-Phosphate Receptor Sorts Lysosomal Hydrolases in the <i>Trans</i> Golgi Network	777
THE TRANSPORT OF MOLECULES BETWEEN THE NUCLEUS AND THE CYTOSOL	735	Defects in the GlcNAc Phosphotransferase Cause a Lysosomal Storage Disease in Humans	779
Nuclear Pore Complexes Perforate the Nuclear Envelope	736	Secretory Vesicles Bud from the <i>Trans</i> Golgi Network	780
Nuclear Localization Signals Direct Proteins to the Nucleus	738	Precursors of Secretory Proteins Are Proteolytically Processed During the Formation of Secretory Vesicles	781
Nuclear Import Receptors Bind to Both Nuclear Localization Signals and NPC Proteins	739	Secretory Vesicles Wait Near the Plasma Membrane Until Signaled to Release Their Contents	782
The Ran GTPase Imposes Directionality on Nuclear Import Through NPCs	740	For Rapid Exocytosis, Synaptic Vesicles Are Primed at the Presynaptic Plasma Membrane	782
Nuclear Export Works Like Nuclear Import, but in Reverse	741	Synaptic Vesicles Can Be Recycled Locally After Exocytosis	783
Transport Through NPCs Can Be Regulated by Controlling Access to the Transport Machinery	742	Secretory Vesicle Membrane Components Are Quickly Removed from the Plasma Membrane	784
The Nuclear Envelope Disassembles and Reassembles During Mitosis	743	Some Regulated Exocytosis Events Serve to Enlarge the Plasma Membrane	785
Summary	745	Polarized Cells Direct Proteins from the <i>Trans</i> Golgi Network to the Appropriate Domain of the Plasma Membrane	786
Problems	746	Summary	787
References	748	TRANSPORT INTO THE CELL FROM THE PLASMA MEMBRANE: ENDOCYTOSIS	788
Chapter 13 Intracellular Membrane Traffic	749	Pinocytic Vesicles Form from Coated Pits in the Plasma Membrane	789
MECHANISMS OF MEMBRANE TRANSPORT AND COMPARTMENT IDENTITY	751	Not All Membrane Invaginations and Pinocytic Vesicles Are Clathrin Coated	789
There Are Various Types of Coated Vesicles	751	Cells Use Receptor-mediated Endocytosis to Import Selected Extracellular Macromolecules	791
		Specific Proteins Are Retrieved from Early Endosomes and Returned to the Plasma Membrane	792
		Recycling Endosomes Regulate Plasma Membrane Composition	793
		Plasma Membrane Signaling Receptors Are Down-regulated by Degradation in Lysosomes	794

Early Endosomes Mature into Late Endosomes	795	CHLOROPLASTS AND PHOTOSYNTHESIS	843
ESCRT Protein Complexes Mediate the Formation of Intraluminal Vesicles in Multivesicular Bodies	796	Chloroplasts Resemble Mitochondria but Have a Separate Thylakoid Compartment	843
Summary	798	Chloroplasts Capture Energy from Sunlight and Use It to Fix Carbon	844
THE DEGRADATION AND RECYCLING OF MACROMOLECULES IN LYSOSOMES	798	Carbon Fixation Uses ATP and NADPH to Convert CO ₂ into Sugars	845
Lysosomes Are the Principal Sites of Intracellular Digestion	798	Carbon Fixation in Some Plants Is Compartmentalized to Facilitate Growth at Low CO ₂ Concentrations	846
Lysosomes Are Heterogeneous	799	The Sugars Generated by Carbon Fixation Can Be Stored as Starch or Consumed to Produce ATP	849
Plant and Fungal Vacuoles Are Remarkably Versatile Lysosomes	800	The Thylakoid Membranes of Chloroplasts Contain the Protein Complexes Required for Photosynthesis and ATP Generation	849
Multiple Pathways Deliver Materials to Lysosomes	801	Chlorophyll-Protein Complexes Can Transfer Either Excitation Energy or Electrons	850
Cells Can Acquire Nutrients from the Extracellular Fluid by Macropinocytosis	802	A Photosystem Contains Chlorophylls in Antennae and a Reaction Center	851
Specialized Phagocytic Cells Can Ingest Large Particles	802	The Thylakoid Membrane Contains Two Different Photosystems Working in Series	852
Cargo Recognition by Cell-surface Receptors Initiates Phagocytosis	803	Photosystem II Uses a Manganese Cluster to Withdraw Electrons from Water	853
Autophagy Degrades Unwanted Proteins and Organelles	804	The Cytochrome <i>b₆-f</i> Complex Connects Photosystem II to Photosystem I	854
The Rate of Nonselective Autophagy Is Regulated by Nutrient Availability	805	Photosystem I Carries Out the Second Charge-Separation Step in the Z Scheme	855
A Family of Cargo-specific Receptors Mediates Selective Autophagy	806	The Chloroplast ATP Synthase Uses the Proton Gradient Generated by the Photosynthetic Light Reactions to Produce ATP	855
Some Lysosomes and Multivesicular Bodies Undergo Exocytosis	807	The Proton-Motive Force for ATP Production in Mitochondria and Chloroplasts Is Essentially the Same	856
Summary	807	Chemiosmotic Mechanisms Evolved in Stages By Providing an Inexhaustible Source of Reducing Power, Photosynthetic Bacteria Overcame a Major Evolutionary Obstacle	857
Problems	808	The Photosynthetic Electron-Transport Chains of Cyanobacteria Produced Atmospheric Oxygen and Permitted New Life-Forms	857
References	810	Summary	860
Chapter 14 Energy Conversion and Metabolic Compartmentation: Mitochondria and Chloroplasts	811	THE GENETIC SYSTEMS OF MITOCHONDRIA AND CHLOROPLASTS	861
THE MITOCHONDRION	813	The Genetic Systems of Mitochondria and Chloroplasts Resemble Those of Prokaryotes	861
The Mitochondrion Has an Outer Membrane and an Inner Membrane	814	Over Time, Mitochondria and Chloroplasts Have Exported Most of Their Genes to the Nucleus by Gene Transfer	862
Fission, Fusion, Distribution, and Degradation of Mitochondria	815	Mitochondria Have a Relaxed Codon Usage and Can Have a Variant Genetic Code	864
The Inner Membrane Cristae Contain the Machinery for Electron Transport and ATP Synthesis	817	Chloroplasts and Bacteria Share Many Striking Similarities	865
The Citric Acid Cycle in the Matrix Produces NADH	817	Organellar Genes Are Maternally Inherited in Animals and Plants	866
Mitochondria Have Many Essential Roles in Cellular Metabolism	818	Mutations in Mitochondrial DNA Can Cause Severe Inherited Diseases	866
A Chemiosmotic Process Couples Oxidation Energy to ATP Production	821	Why Do Mitochondria and Chloroplasts Maintain a Costly Separate System for DNA Transcription and Translation?	867
The Energy Derived from Oxidation Is Stored as an Electrochemical Gradient	822	Summary	868
Summary	823	Problems	869
THE PROTON PUMPS OF THE ELECTRON-TRANSPORT CHAIN	823	References	871
The Redox Potential Is a Measure of Electron Affinities	823	Chapter 15 Cell Signaling	873
Electron Transfers Release Large Amounts of Energy	824	PRINCIPLES OF CELL SIGNALING	873
Transition Metal Ions and Quinones Accept and Release Electrons Readily	824	Extracellular Signals Can Act Over Short or Long Distances	874
NADH Transfers Its Electrons to Oxygen Through Three Large Enzyme Complexes Embedded in the Inner Membrane	827	Extracellular Signal Molecules Bind to Specific Receptors	875
The NADH Dehydrogenase Complex Contains Separate Modules for Electron Transport and Proton Pumping	828	Each Cell Is Programmed to Respond to Specific Combinations of Extracellular Signals	876
Cytochrome <i>c</i> Reductase Takes Up and Releases Protons on Opposite Sides of the Crista Membrane, Thereby Pumping Protons	829	There Are Three Major Classes of Cell-Surface Receptor Proteins	878
The Cytochrome <i>c</i> Oxidase Complex <i>Pumps</i> Protons and Reduces O ₂ Using a Catalytic Iron-Copper Center	831	Cell-Surface Receptors Relay Signals Via Intracellular Signaling Molecules	879
Succinate Dehydrogenase Acts in Both the Electron-Transport Chain and the Citric Acid Cycle	832	Intracellular Signals Must Be Specific and Robust in a Noisy Cytoplasm	881
The Respiratory Chain Forms a Supercomplex in the Crista Membrane	833	Intracellular Signaling Complexes Form at Activated Cell-Surface Receptors	882
Protons Can Move Rapidly Through Proteins Along Predefined Pathways	834	Modular Interaction Domains Mediate Interactions Between Intracellular Signaling Proteins	883
Summary	835	The Relationship Between Signal and Response Varies in Different Signaling Pathways	885
ATP PRODUCTION IN MITOCHONDRIA	835	The Speed of a Response Depends on the Turnover of Signaling Molecules	886
The Large Negative Value of ΔG for ATP Hydrolysis Makes ATP Useful to the Cell	835		
The ATP Synthase Is a Nanomachine That Produces ATP by Rotary Catalysis	837		
Proton-driven Turbines Are Ancient and Critical for Energy Conversion	839		
Mitochondrial Cristae Help to Make ATP Synthesis Efficient	840		
Special Transport Proteins Move Solutes Through the Inner Membrane	841		
Chemiosmotic Mechanisms First Arose in Bacteria	842		
Summary	842		

Cells Can Respond Abruptly to a Gradually Increasing Signal	887	Chapter 16 The Cytoskeleton	949
Positive Feedback Can Generate an All-or-None Response	888	FUNCTION AND DYNAMICS OF THE CYTOSKELETON	949
Negative Feedback Is a Common Feature of Intracellular Signaling Systems	890	Cytoskeletal Filaments Are Dynamic, but Can Nevertheless	
Cells Can Adjust Their Sensitivity to a Signal	890	Form Stable Structures	951
Summary	892	The Cytoskeleton Determines Cellular Organization and Polarity	952
SIGNALING THROUGH G-PROTEIN-COUPLED RECEPTORS	892	Filaments Assemble from Protein Subunits That Impart	
Heterotrimeric G Proteins Relay Signals from GPCRs	893	Specific Physical and Dynamic Properties	953
Some G Proteins Regulate the Production of Cyclic AMP	895	Accessory Proteins and Motors Act on Cytoskeletal Filaments	955
Cyclic-AMP-dependent Protein Kinase (PKA) Mediates Most of the Effects of Cyclic AMP	896	Molecular Motors Operate in a Cellular Environment	
Some G Proteins Signal Via Phospholipids	898	Dominated by Brownian Motion	956
Ca ²⁺ Functions as a Ubiquitous Intracellular Mediator	899	Summary	957
Feedback Generates Ca ²⁺ Waves and Oscillations	900	ACTIN	957
Ca ²⁺ /Calmodulin-dependent Protein Kinases Mediate Many Responses to Ca ²⁺ Signals	902	Actin Subunits Assemble Head-to-Tail to Create Flexible,	
Some G Proteins Directly Regulate Ion Channels	904	Polar Filaments	958
Smell and Vision Depend on GPCRs That Regulate Ion Channels	905	Nucleation Is the Rate-limiting Step in the Formation of	
Nitric Oxide Gas Can Mediate Signaling Between Cells	908	Actin Filaments	958
Second Messengers and Enzymatic Cascades Amplify Signals	909	Actin Filaments Have Two Distinct Ends That Grow at Different Rates	962
GPCR Desensitization Depends on Receptor Phosphorylation	909	ATP Hydrolysis Within Actin Filaments Leads to Treadmilling	
Summary	910	at Steady State	962
SIGNALING THROUGH ENZYME-COUPLED RECEPTORS	911	The Functions of Actin Filaments Are Inhibited by Both	
Activated Receptor Tyrosine Kinases (RTKs) Phosphorylate Themselves	911	Polymer-stabilizing and Polymer-destabilizing Chemicals	963
Phosphorylated Tyrosines on RTKs Serve as Docking Sites for Intracellular Signaling Proteins	913	Actin-binding Proteins Influence Filament Dynamics	
Proteins with SH2 Domains Bind to Phosphorylated Tyrosines	913	and Organization	964
The Monomeric GTPase Ras Mediates Signaling by Most RTKs	915	Actin Nucleation Is Tightly Regulated and Generates Branched or Straight Filaments	964
Ras Activates a MAP Kinase Signaling Module	916	Actin Filament Elongation Is Regulated by Monomer-binding Proteins	967
Scaffold Proteins Reduce Cross-Talk Between Different MAP Kinase Modules	918	Actin Filament-binding Proteins Alter Filament Dynamics and Organization	968
Rho Family GTPases Functionally Couple Cell-Surface Receptors to the Cytoskeleton	919	Severing Proteins Regulate Actin Filament Depolymerization	970
PI 3-Kinase Produces Lipid Docking Sites in the Plasma Membrane	920	Bacteria Can Hijack the Host Actin Cytoskeleton	971
The PI-3-Kinase-Akt Signaling Pathway Stimulates Animal Cells to Survive and Grow	921	Actin at the Cell Cortex Determines Cell Shape	971
RTKs and GPCRs Activate Overlapping Signaling Pathways	923	Distinct Modes of Cell Migration Rely on the Actin Cytoskeleton	972
Some Enzyme-coupled Receptors Associate with Cytoplasmic Tyrosine Kinases	923	Cells Migrating in Three Dimensions Can Navigate Around Barriers	974
Cytokine Receptors Activate the JAK-STAT Signaling Pathway	924	Summary	975
Extracellular Signal Proteins of the TGF β Superfamily Act Through Receptor Serine/Threonine Kinases and Smads	926	MYOSIN AND ACTIN	976
Summary	927	Actin-based Motor Proteins Are Members of the	
ALTERNATIVE SIGNALING ROUTES IN GENE REGULATION	928	Myosin Superfamily	976
The Receptor Notch Is a Latent Transcription Regulator	928	Myosin Generates Force by Coupling ATP Hydrolysis	
Wnt Proteins Activate Frizzled and Thereby Inhibit β -Catenin Degradation	930	to Conformational Changes	977
Hedgehog Proteins Initiate a Complex Signaling Pathway in the Primary Cilium	932	Sliding of Myosin II Along Actin Filaments Causes Muscles to Contract	977
Many Inflammatory and Stress Signals Act Through an NF κ B-dependent Signaling Pathway	934	A Sudden Rise in Cytosolic Ca ²⁺ Concentration Initiates	
Nuclear Receptors Are Ligand-modulated Transcription Regulators	935	Muscle Contraction	981
Circadian Clocks Use Negative Feedback Loops to Control Gene Expression	937	Heart Muscle Is a Precisely Engineered Machine	984
Three Purified Proteins Can Reconstitute a Cyanobacterial Circadian Clock in a Test Tube	938	Actin and Myosin Perform a Variety of Functions in Non-Muscle Cells	984
Summary	939	Summary	986
SIGNALING IN PLANTS	940	MICROTUBULES	987
Multicellularity and Cell Communication Evolved Independently in Plants and Animals	940	Microtubules Are Hollow Tubes Made of Protofilaments	988
Receptor Serine/Threonine Kinases Are the Largest Class of Cell-Surface Receptors in Plants	941	Microtubules Undergo a Process Called Dynamic Instability	988
Ethylene Blocks the Degradation of Specific Transcription Regulatory Proteins in the Nucleus	941	Microtubule Functions Are Inhibited by Both Polymer-stabilizing and Polymer-destabilizing Drugs	991
Regulated Positioning of Auxin Transporters Patterns Plant Growth	943	A Protein Complex Containing γ -Tubulin Nucleates Microtubules	991
Phytochromes Detect Red Light, and Cryptochromes Detect Blue Light	944	The Centrosome Is a Prominent Microtubule Nucleation Site	991
Summary	945	Microtubule Organization Varies Widely Among Cell Types	993
Problems	946	Microtubule-binding Proteins Modulate Filament Dynamics and Organization	995
References	948	Microtubule Plus End-binding Proteins Modulate Microtubule Dynamics and Attachments	996
		Tubulin-sequestering and Microtubule-severing Proteins Modulate Microtubule Dynamics	998
		Two Types of Motor Proteins Move Along Microtubules	999
		Microtubules and Motors Move Organelles and Vesicles	1002
		Motile Cilia and Flagella Are Built from Microtubules and Dyneins	1004
		Primary Cilia Perform Important Signaling Functions in Animal Cells	1005
		Summary	1006
		INTERMEDIATE FILAMENTS AND OTHER CYTOSKELETAL POLYMERS	1007
		Intermediate Filament Structure Depends on the Lateral Bundling and Twisting of Coiled-Coils	1007
		Intermediate Filaments Impart Mechanical Stability to Animal Cells	1009
		Linker Proteins Connect Cytoskeletal Filaments and Bridge the Nuclear Envelope	1011

Septins Form Filaments That Contribute to Subcellular Organization	1012	Actin and Myosin II in the Contractile Ring Guide the Process of Cytokinesis	1065
Bacterial Cell Shape and Division Depend on Homologs of Eukaryotic Cytoskeletal Proteins	1013	Local Activation of RhoA Triggers Assembly and Contraction of the Contractile Ring	1065
Summary	1016	The Microtubules of the Mitotic Spindle Determine the Plane of Animal Cell Division	1066
CELL POLARITY AND COORDINATION OF THE CYTOSKELETON	1016	The Phragmoplast Guides Cytokinesis in Higher Plants	1068
Cell Polarity Is Governed by Small GTPases in Budding Yeast	1016	Membrane-enclosed Organelles Must Be Distributed to Daughter Cells During Cytokinesis	1069
PAR Proteins Generate Anterior–Posterior Polarity in Embryos	1018	Some Cells Reposition Their Spindle to Divide Asymmetrically	1069
Conserved Complexes Polarize Epithelial Cells and Control Their Growth	1019	Mitosis Can Occur Without Cytokinesis	1070
Cell Migration Requires Dynamic Cell Polarity	1020	Summary	1070
External Signals Can Dictate the Direction of Cell Migration	1022	MEIOSIS	1071
Communication Among Cytoskeletal Elements Supports Whole-Cell Polarity and Locomotion	1023	Meiosis Includes Two Rounds of Chromosome Segregation	1071
Summary	1023	Duplicated Homologs Pair During Meiotic Prophase	1073
Problems	1024	Homolog Pairing Culminates in the Formation of a Synaptonemal Complex	1073
References	1025	Homolog Segregation Depends on Several Unique Features of Meiosis I	1075
Chapter 17 The Cell Cycle	1027	Crossing-Over Is Highly Regulated	1076
OVERVIEW OF THE CELL CYCLE	1027	Meiosis Frequently Goes Wrong	1077
The Eukaryotic Cell Cycle Usually Consists of Four Phases	1028	Summary	1077
Cell-Cycle Control Is Similar in All Eukaryotes	1030	CONTROL OF CELL DIVISION AND CELL GROWTH	1077
Cell-Cycle Progression Can Be Studied in Various Ways	1030	Mitogens Stimulate Cell Division	1078
Summary	1031	Cells Can Enter a Specialized Nondividing State	1078
THE CELL-CYCLE CONTROL SYSTEM	1031	Mitogens Stimulate G ₁ -Cdk and G ₁ /S-Cdk Activities	1079
The Cell-Cycle Control System Triggers the Major Events of the Cell Cycle	1032	DNA Damage Blocks Cell Division	1080
The Cell-Cycle Control System Depends on Cyclically Activated Cyclin-dependent Protein Kinases	1033	Many Human Cells Have a Built-In Limitation on the Number of Times They Can Divide	1082
Protein Phosphatases Reverse the Effects of Cdks	1035	Cell Proliferation Is Accompanied by Cell Growth	1083
Hundreds of Cdk Substrates Are Phosphorylated in a Defined Order	1035	Proliferating Cells Usually Coordinate Their Growth and Division	1084
Positive Feedback Generates the Switchlike Behavior of Cell-Cycle Transitions	1036	Summary	1084
The Anaphase-promoting Complex/Cyclosome (APC/C) Triggers the Metaphase-to-Anaphase Transition	1038	Problems	1085
The G ₁ Phase Is a Stable State of Cdk Inactivity	1040	References	1087
The Cell-Cycle Control System Functions as a Linked Series of Biochemical Switches	1041	Chapter 18 Cell Death	1089
Summary	1042	Apoptosis Eliminates Unwanted Cells	1090
S PHASE	1042	Apoptosis Depends on an Intracellular Proteolytic Cascade Mediated by Caspases	1091
S-Cdk Initiates DNA Replication Once Per Cell Cycle	1043	Activation of Cell-Surface Death Receptors Initiates the Extrinsic Pathway of Apoptosis	1093
Chromosome Duplication Requires Duplication of Chromatin Structure	1045	The Intrinsic Pathway of Apoptosis Depends on Proteins Released from Mitochondria	1094
Cohesins Hold Sister Chromatids Together	1045	Bcl2 Proteins Are the Critical Controllers of the Intrinsic Pathway of Apoptosis	1095
Summary	1046	An Inhibitor of Apoptosis (an IAP) and Two Anti-IAP Proteins Help Control Caspase Activation in the Cytosol of Some Mammalian Cells	1098
MITOSIS	1046	Extracellular Survival Factors Inhibit Apoptosis in Various Ways	1098
M-Cdk and Other Protein Kinases Drive Entry into Mitosis	1047	Healthy Neighbors Phagocytose and Digest Apoptotic Cells	1100
Condensin Helps Configure Duplicated Chromosomes for Separation	1047	Either Excessive or Insufficient Apoptosis Can Contribute to Disease	1100
The Mitotic Spindle Is a Dynamic Microtubule-based Machine	1050	Summary	1102
Microtubules Are Nucleated in Multiple Regions of the Spindle	1051	Problems	1103
Microtubule Instability Increases Greatly in Mitosis	1052	References	1104
Microtubule-based Motor Proteins Govern Spindle Assembly and Function	1052	Chapter 19 Cell Junctions and the Extracellular Matrix	1105
Bipolar Spindle Assembly in Most Animal Cells Begins with Centrosome Duplication	1053	CELL–CELL JUNCTIONS	1108
Spindle Assembly in Animal Cells Requires Nuclear-Envelope Breakdown	1054	Cadherins Form a Diverse Family of Adhesion Molecules	1108
Mitotic Chromosomes Promote Bipolar Spindle Assembly	1055	Cadherins Mediate Homophilic Adhesion	1108
Kinetochores Attach Sister Chromatids to the Spindle	1056	Cadherin-dependent Cell–Cell Adhesion Guides the Organization of Developing Tissues	1110
Bi-orientation Is Achieved by Trial and Error	1057	Assembly of Strong Cell–Cell Adhesions Requires Changes in the Actin Cytoskeleton	1112
Multiple Forces Act on Chromosomes in the Spindle	1059	Catenins Link Classical Cadherins to the Actin Cytoskeleton	1113
The APC/C Triggers Sister-Chromatid Separation and the Completion of Mitosis	1060	Adherens Junctions Respond to Tension from Inside and Outside the Tissue	1113
Unattached Chromosomes Block Sister-Chromatid Separation: The Spindle Assembly Checkpoint	1062	Tissue Remodeling Depends on the Coordination of Actin-mediated Contraction with Cell–Cell Adhesion	1114
Chromosomes Segregate in Anaphase A and B	1062	Desmosomes Give Epithelia Mechanical Strength	1116
Segregated Chromosomes Are Packaged in Daughter Nuclei at Telophase	1063	Tight Junctions Form a Seal Between Cells and a Fence Between Plasma Membrane Domains	1116
Summary	1064		
CYTOKINESIS	1064		

Tight Junctions Contain Strands of Transmembrane Adhesion Proteins	1119	Cancer Cells Contain Somatic Mutations	1166
Scaffold Proteins Organize Junctional Protein Complexes	1120	A Single Mutation Is Not Enough to Change a Normal Cell into a Cancer Cell	1166
Gap Junctions Couple Cells Both Electrically and Metabolically	1121	Many Cancers Develop Gradually Through Successive Rounds of Random Inherited Change Followed by Natural Selection	1167
A Gap-Junction Connexon Is Made of Six Transmembrane Connexin Subunits	1122	Cancers Can Evolve Abruptly Due to Genetic Instability	1168
In Plants, Plasmodesmata Perform Many of the Same Functions as Gap Junctions	1123	Some Cancers May Harbor a Small Population of Stem Cells	1170
Selectins Mediate Transient Cell-Cell Adhesions in the Bloodstream	1125	A Common Set of Hallmarks Typically Characterizes Cancerous Growth	1171
Members of the Immunoglobulin Superfamily Mediate Ca ²⁺ -independent Cell-Cell Adhesion	1126	Cancer Cells Display an Altered Control of Growth and Homeostasis	1172
Summary	1127	Human Cancer Cells Escape a Built-in Limit to Cell Proliferation	1173
THE EXTRACELLULAR MATRIX OF ANIMALS	1127	Cancer Cells Have an Abnormal Ability to Bypass Death Signals	1174
The Extracellular Matrix Is Made and Oriented by the Cells Within It	1128	Cancer Cells Have Altered Sugar Metabolism	1175
Glycosaminoglycan (GAG) Chains Occupy Large Amounts of Space and Form Hydrated Gels	1129	The Tumor Microenvironment Influences Cancer Development	1175
Hyaluronan Acts as a Space Filler During Tissue Morphogenesis and Repair	1129	Cancer Cells Must Survive and Proliferate in a Foreign Environment	1176
Proteoglycans Are Composed of GAG Chains Covalently Linked to a Core Protein	1130	Summary	1178
Collagens Are the Major Proteins of the Extracellular Matrix	1132	CANCER-CRITICAL GENES: HOW THEY ARE FOUND AND WHAT THEY DO	1178
Collagen Chains Undergo a Series of Post-translational Modifications	1133	The Identification of Gain-of-Function and Loss-of-Function Cancer Mutations Has Traditionally Required Different Methods	1179
Secreted Fibril-associated Collagens Help Organize the Fibrils	1135	Retroviruses Led to the Identification of Oncogenes	1180
Elastin Gives Tissues Their Elasticity	1136	Genes Mutated in Cancer Can Be Made Overactive in Many Ways	1181
Cells Govern and Respond to the Mechanical Properties of the Matrix	1137	Studies of Rare Hereditary Cancer Syndromes First Identified Tumor Suppressor Genes	1182
Fibronectin and Other Multidomain Glycoproteins Help Organize the Matrix	1138	Both Genetic and Epigenetic Mechanisms Can Inactivate Tumor Suppressor Genes	1183
Fibronectin Binds to Integrins	1139	Systematic Sequencing of Cancer Cell Genomes Has Transformed Our Understanding of the Disease	1184
Tension Exerted by Cells Regulates the Assembly of Fibronectin Fibrils	1140	Many Cancers Have an Extraordinarily Disrupted Genome	1185
The Basal Lamina Is a Specialized Form of Extracellular Matrix	1141	Epigenetic and Chromatin Changes Contribute to Most Cancers	1185
Laminin and Type IV Collagen Are Major Components of the Basal Lamina	1143	Hundreds of Human Genes Contribute to Cancer	1186
Basal Laminae Have Diverse Functions	1144	Disruptions in a Handful of Key Pathways Are Common to Many Cancers	1187
Cells Have to Be Able to Degrade Matrix, as Well as Make It	1145	Mutations in the PI 3-kinase/Akt/mTOR Pathway Drive Cancer Cells to Grow	1188
Matrix Proteoglycans and Glycoproteins Regulate the Activities of Secreted Proteins	1146	Mutations in the p53 Pathway Enable Cancer Cells to Survive and Proliferate Despite Stress and DNA Damage	1189
Summary	1147	Studies Using Mice Help to Define the Functions of Cancer-critical Genes	1190
CELL-MATRIX JUNCTIONS	1147	Cancers Become More and More Heterogeneous as They Progress	1192
Integrins Are Transmembrane Heterodimers That Link the Extracellular Matrix to the Cytoskeleton	1147	Colorectal Cancers Evolve Slowly Via a Succession of Visible Changes	1192
Integrin Defects Are Responsible for Many Genetic Diseases	1148	A Few Key Genetic Lesions Are Common to a Large Fraction of Colorectal Cancers	1194
Integrins Can Switch Between an Active and an Inactive Conformation	1149	Some Colorectal Cancers Have Defects in DNA Mismatch Repair	1195
Integrins Cluster to Form Strong Adhesions	1151	The Steps of Tumor Progression Can Often Be Correlated with Specific Mutations	1196
Extracellular Matrix Attachments Act Through Integrins to Control Cell Proliferation and Survival	1151	The Changes in Tumor Cells That Lead to Metastasis Are Still Largely a Mystery	1197
Integrins Recruit Intracellular Signaling Proteins at Sites of Cell-Matrix Adhesion	1152	Summary	1197
Cell-Matrix Adhesions Respond to Mechanical Forces	1153	CANCER PREVENTION AND TREATMENT: PRESENT AND FUTURE	1198
Summary	1154	Epidemiology Reveals That Many Cases of Cancer Are Preventable	1198
THE PLANT CELL WALL	1154	Sensitive Assays Can Detect Those Cancer-causing Agents That Damage DNA	1199
The Composition of the Cell Wall Depends on the Cell Type	1155	Fifty Percent of Cancers Could Be Prevented by Changes in Lifestyle	1200
The Tensile Strength of the Cell Wall Allows Plant Cells to Develop Turgor Pressure	1155	Viruses and Other Infections Contribute to a Significant Proportion of Human Cancers	1201
The Primary Cell Wall Is Built from Cellulose Microfibrils Interwoven with a Network of Pectic Polysaccharides	1156	Cancers of the Uterine Cervix Can Be Prevented by Vaccination Against Human Papillomavirus	1202
Oriented Cell Wall Deposition Controls Plant Cell Growth	1157	Infectious Agents Can Cause Cancer in a Variety of Ways	1203
Microtubules Orient Cell Wall Deposition	1158	The Search for Cancer Cures Is Difficult but Not Hopeless	1204
Summary	1159	Traditional Therapies Exploit the Genetic Instability and Loss of Cell-Cycle Checkpoint Responses in Cancer Cells	1204
Problems	1160	New Drugs Can Kill Cancer Cells Selectively by Targeting Specific Mutations	1204
References	1162	PARP Inhibitors Kill Cancer Cells That Have Defects in <i>Brca1</i> or <i>Brca2</i> Genes	1205
Chapter 20 Cancer	1163		
CANCER AS A MICROEVOLUTIONARY PROCESS	1163		
Cancer Cells Bypass Normal Proliferation Controls and Colonize Other Tissues	1164		
Most Cancers Derive from a Single Abnormal Cell	1165		

Small Molecules Can Be Designed to Inhibit Specific Oncogenic Proteins	1207	A Gene Expression Oscillator Acts as a Clock to Control Vertebrate Segmentation	1249
Many Cancers May Be Treatable by Enhancing Immune Responses	1209	Cell-intrinsic Timing Mechanisms Can Lead to Different Cell Fates	1251
Immunosuppression Is a Major Hurdle for Cancer Immunotherapy	1210	Cells Rarely Count Cell Divisions to Time Their Development	1252
Cancers Evolve Resistance to Therapies	1212	MicroRNAs Can Regulate Developmental Transitions	1252
We Now Have the Tools to Devise Combination Therapies Tailored to the Individual	1212	Cell and Nuclear Size Relationships Schedule the Onset of Zygotic Gene Expression	1254
Summary	1213	Hormonal Signals Coordinate the Timing of Developmental Transitions	1255
Problems	1214	Environmental Cues Determine the Time of Flowering	1256
References	1216	Summary	1257
Chapter 21 Development of Multicellular Organisms	1217	MORPHOGENESIS	1257
OVERVIEW OF DEVELOPMENT	1218	Imbalance in Physical Forces Acting on Cells Drives Morphogenesis	1258
Conserved Mechanisms Establish the Core Tissues of Animals	1218	Tension and Adhesion Determine Cell Packing Within Epithelial Sheets	1258
The Developmental Potential of Cells Becomes Progressively Restricted	1219	Changing Patterns of Cell Adhesion Molecules Force Cells into New Arrangements	1259
Cell Memory Underlies Cell Decision-Making	1220	Repulsive Interactions Help Maintain Tissue Boundaries	1259
Several Model Organisms Have Been Crucial for Understanding Development	1220	Groups of Similar Cells Can Perform Dramatic Collective Rearrangements	1261
Regulatory DNA Seems Largely Responsible for the Differences Between Animal Species	1220	Planar Cell Polarity Orients Cell Behaviors Within an Embryo	1261
Small Numbers of Conserved Cell–Cell Signaling Pathways Coordinate Spatial Patterning	1221	An Epithelium Can Bend During Development to Form a Tube	1263
Through Combinatorial Control and Cell Memory, Simple Signals Can Generate Complex Patterns	1221	Interactions Between an Epithelium and Mesenchyme Generate Branching Tubular Structures	1264
Morphogens Are Diffusible Inductive Signals That Exert Graded Effects	1222	The Extracellular Matrix Also Influences Tissue Shape	1265
Lateral Inhibition Can Generate Patterns of Different Cell Types	1223	Cell Migration Is Guided by Environmental Signals	1266
Asymmetric Cell Division Can Also Generate Diversity	1224	The Distribution of Migrant Cells Depends on Survival Factors	1267
Initial Patterns Are Established in Small Fields of Cells and Refined by Sequential Induction as the Embryo Grows	1225	Cells Migrate in Groups to Achieve Large-Scale Morphogenetic Movements	1268
Developmental Biology Provides Insights into Disease and Tissue Maintenance	1225	Summary	1269
Summary	1226	GROWTH	1269
MECHANISMS OF PATTERN FORMATION	1226	The Proliferation, Death, and Size of Cells Determine Organ and Organism Size	1270
Different Animals Use Different Mechanisms to Establish Their Primary Axes of Polarization	1226	Changes in Cell Size Usually Result from Modified Cell Cycles	1271
Studies in <i>Drosophila</i> Have Revealed Many Genetic Control Mechanisms Underlying Development	1228	Animals and Organs Can Assess and Regulate Total Cell Mass	1272
Gene Products Deposited in the Egg Organize the Axes of the Early <i>Drosophila</i> Embryo	1228	Various Extracellular Signals Stimulate or Inhibit Growth	1273
Three Groups of Genes Control <i>Drosophila</i> Segmentation Along the A-P Axis	1230	The Hippo Pathway Relays Mechanical Signals Regulating Growth	1273
A Hierarchy of Gene Regulatory Interactions Subdivides the <i>Drosophila</i> Embryo	1231	Hormones Coordinate Growth Throughout the Body	1274
Egg-Polarity, Gap, and Pair-Rule Genes Create a Transient Pattern That Is Remembered by Segment-Polarity and <i>Hox</i> Genes	1233	The Duration of Growth Influences Organism Size	1275
<i>Hox</i> Genes Permanently Pattern the A-P Axis	1233	Summary	1275
<i>Hox</i> Proteins Give Each Segment Its Individuality	1234	Problems	1276
<i>Hox</i> Genes Are Expressed According to Their Order in the <i>Hox</i> Complex	1234	References	1278
Trithorax and Polycomb Group Proteins Regulate <i>Hox</i> Expression to Maintain a Permanent Record of Positional Information	1235	Chapter 22 Stem Cells in Tissue Homeostasis and Regeneration	1279
The D-V Signaling Genes Create a Gradient of the Transcription Regulator Dorsal	1236	STEM CELLS AND TISSUE HOMEOSTASIS	1279
A Hierarchy of Inductive Interactions Subdivides the Vertebrate Embryo	1238	Stem Cells Are Defined by Their Ability to Self-renew and Produce Differentiated Cells	1280
A Competition Between Secreted Signaling Proteins Patterns the Vertebrate Embryonic Axes	1239	The Epithelial Lining of the Small Intestine Is Continually Renewed Through Cell Proliferation in Crypts	1281
<i>Hox</i> Genes Control the Vertebrate A-P Axis	1240	Epidermal Stem Cells Maintain a Self-renewing, Waterproof, Epithelial Barrier on the Body Surface	1282
Some Transcription Regulators Can Activate a Program That Defines a Cell Type or Creates an Entire Organ	1241	Cell Lineage Tracing Reveals the Location of Stem Cells and Their Progeny	1284
Notch-mediated Lateral Inhibition Refines Cellular Spacing Patterns	1242	Quiescent Stem Cells Are Difficult to Identify by Lineage Tracing	1285
Cell-fate Determinants Can Be Asymmetrically Inherited	1244	Hematopoietic Stem Cells Can Be Identified by Transplantation	1286
Evolution of Regulatory DNA Explains Many Morphological Differences	1245	Some Tissues Do Not Require Stem Cells for Their Maintenance	1289
Summary	1247	In Response to Injury, Some Differentiated Cells Can Revert to Progenitor Cells and Some Progenitor Cells Can Revert to Stem Cells	1289
DEVELOPMENTAL TIMING	1248	Some Tissues Lack Stem Cells and Are Not Renewable	1290
Molecular Lifetimes Play a Critical Part in Developmental Timing	1248	Summary	1290
		CONTROL OF STEM-CELL FATE AND SELF-RENEWAL	1291
		The Stem-Cell Niche Maintains Stem-Cell Self-Renewal	1291
		The Size of the Niche Can Determine the Number of Stem Cells	1292
		Asymmetric Stem-Cell Division Can Maintain Stem-Cell Number	1293
		In Many Symmetric Stem-Cell Divisions, Daughter Cells Choose Their Fates Independently and Stochastically	1294
		A Decline in Stem-Cell Function Contributes to Tissue Aging	1294

Summary	1296
REGENERATION AND REPAIR	1296
Planarian Flatworms Contain Stem Cells That Can Regenerate a Whole New Body	1297
Some Vertebrates Can Regenerate Entire Limbs and Organs	1298
Stem Cells Can Be Used Clinically to Replace Lost Hematopoietic or Skin Cells	1299
Neural Stem Cells Can Be Manipulated in Culture and Used to Repopulate a Diseased Central Nervous System	1299
Summary	1300
CELL REPROGRAMMING AND PLURIPOTENT STEM CELLS	1300
Nuclei Can Be Reprogrammed by Transplantation into Foreign Cytoplasm	1301
Reprogramming of a Transplanted Nucleus Involves Drastic Changes in Chromatin	1301
Embryonic Stem (ES) Cells Can Generate Any Part of the Body	1302
A Core Set of Transcription Regulators Defines and Maintains the ES-Cell State	1303
Fibroblasts Can Be Reprogrammed to Create Induced Pluripotent Stem (iPS) Cells	1303
Reprogramming Involves a Massive Upheaval of the Gene Control System	1304
An Experimental Manipulation of Factors That Modify Chromatin Can Increase Reprogramming Efficiencies	1305
ES and iPS Cells Can Be Guided to Generate Specific Adult Cell Types and Even Organoids	1306
Cells of One Specialized Type Can Be Forced to Transdifferentiate Directly into Another	1306
ES and iPS Cells Are Also Useful for Drug Discovery and Analysis of Disease	1308
Summary	1309
Problems	1310
References	1312

Chapter 23 Pathogens and Infection 1313

INTRODUCTION TO PATHOGENS	1313
Pathogens Can Be Viruses, Bacteria, or Eukaryotes	1314
Pathogens Interact with Their Hosts in Different Ways	1314
Bacteria Are Diverse and Occupy a Remarkable Variety of Ecological Niches	1315
Bacterial Pathogens Carry Specialized Virulence Genes	1317
Bacterial Virulence Genes Encode Toxins and Secretion Systems That Deliver Effector Proteins to Host Cells	1319
Fungal and Protozoan Parasites Have Complex Life Cycles Involving Multiple Forms	1321
All Aspects of Viral Propagation Depend on Host-Cell Machinery	1322
Summary	1325
CELL BIOLOGY OF PATHOGEN INFECTION	1325
Pathogens Breach Epithelial Barriers to Infect the Host	1326
Pathogens That Colonize an Epithelium Must Overcome Its Protective Mechanisms	1326
Extracellular Pathogens Use Toxins and Contact-dependent Secretion Systems to Disturb Host Cells Without Entering Them	1328
Intracellular Pathogens Have Mechanisms for Both Entering and Leaving Host Cells	1329
Viruses Bind to Virus Receptors at the Host-Cell Surface	1329
Viruses Enter Host Cells by Membrane Fusion, Pore Formation, or Membrane Disruption	1330
Bacteria Enter Host Cells by Phagocytosis	1331
Intracellular Eukaryotic Parasites Actively Invade Host Cells	1333
Some Intracellular Pathogens Escape from the Phagosome into the Cytosol	1334
Many Pathogens Alter Membrane Traffic in the Host Cell to Survive and Replicate	1335
Bacteria and Viruses Use the Host-Cell Cytoskeleton for Intracellular Movement	1338
Many Microbes Manipulate Autophagy	1340
Viruses Can Take Over the Metabolism of the Host Cell	1340
Pathogens Can Evolve Rapidly by Antigenic Variation	1341
Error-prone Replication Dominates Viral Evolution	1343
Drug-resistant Pathogens Are a Growing Problem	1344
Summary	1346

THE HUMAN MICROBIOTA	1347
The Human Microbiota Is a Complex Ecological System	1347
The Microbiota Influences Our Development and Health	1348
Summary	1349
Problems	1350
References	1351

Chapter 24 The Innate and Adaptive Immune Systems 1353

THE INNATE IMMUNE SYSTEM	1354
Epithelial Surfaces Serve as Barriers to Infection	1354
Pattern Recognition Receptors (PRRs) Recognize Conserved Features of Pathogens	1354
There Are Multiple Families of PRRs	1355
Activated PRRs Trigger an Inflammatory Response at Sites of Infection	1356
Phagocytic Cells Seek, Engulf, and Destroy Pathogens	1358
Complement Activation Targets Pathogens for Phagocytosis or Lysis	1358
Virus-infected Cells Take Drastic Measures to Prevent Viral Replication	1360
Natural Killer Cells Induce Virus-infected Cells to Kill Themselves	1361
Dendritic Cells Provide the Link Between the Innate and Adaptive Immune Systems	1362
Summary	1362
OVERVIEW OF THE ADAPTIVE IMMUNE SYSTEM	1364
B Cells Develop in the Bone Marrow, T Cells in the Thymus	1365
Immunological Memory Depends on Both Clonal Expansion and Lymphocyte Differentiation	1366
Most B and T Cells Continually Recirculate Through Peripheral Lymphoid Organs	1368
Immunological Self-tolerance Ensures That B and T Cells Do Not Attack Normal Host Cells and Molecules	1370
Summary	1372
B CELLS AND IMMUNOGLOBULINS	1372
B Cells Make Immunoglobulins (Igs) as Both Cell-Surface Antigen Receptors and Secreted Antibodies	1373
Mammals Make Five Classes of Igs	1373
Ig Light and Heavy Chains of Antibodies Consist of Constant and Variable Regions	1375
Ig Genes Are Assembled from Separate Gene Segments During B Cell Development	1377
Antigen-driven Somatic Hypermutation Fine-Tunes Antibody Responses	1379
B Cells Can Switch the Class of Ig They Make	1379
Summary	1381
T CELLS AND MHC PROTEINS	1382
T Cell Receptors (TCRs) Are Ig-like Heterodimers	1382
Activated Dendritic Cells Activate Naïve T Cells	1383
T Cells Recognize Foreign Peptides Bound to MHC Proteins	1384
MHC Proteins Are the Most Polymorphic Human Proteins Known	1388
CD4 and CD8 Co-receptors on T Cells Bind to Invariant Parts of MHC Proteins	1389
Developing Thymocytes Undergo Positive and Negative Selection	1389
Cytotoxic T Cells Induce Infected Target Cells to Undergo Apoptosis	1391
Effector Helper T Cells Help Activate Other Cells of the Innate and Adaptive Immune Systems	1392
Naïve Helper T Cells Can Differentiate into Different Types of Effector T Cells	1393
Both T and B Cells Require Multiple Extracellular Signals for Activation	1394
Many Cell-Surface Proteins Belong to the Ig Superfamily	1396
Vaccination Against Pathogens Has Been Immunology's Greatest Contribution to Human Health	1396
Summary	1400
Problems	1402
References	1404

Glossary G:1

Index I:1